

BY ODHIAMBO DARREN GREGORY



KENYATTA UNIVERSITY

**DEPARTMENT OF EDUCATIONAL COMMUNICATION AND
TECHNOLOGY**

**ECT 211: CONTEMPORARY PEDAGOGIES IN TEACHER
EDUCATION**

GENERAL INTRODUCTION

Teacher education is the hub of professional development of teachers. Pedagogy is the crucial cog in the wheel of this process. Teacher preparation focuses, *inter alia*, on equipping prospective teachers with requisite knowledge, skills and attitudes in order for them to discharge their duties with due diligence and commitment.

This module on contemporary pedagogies in teacher education seeks to play a significant role in preparing pre-service teachers to enable them deliver acceptable services in schools and related educational institutions. While it does not by any means serve as a *panacea* to the myriad of issues in the field of education today, it however *does* serve as an ‘eye opener’ in terms of instructional planning and implementation of curriculum.

The module consists of twelve **lessons** which collectively provide ideas on how to prepare for teaching with specific learning outcomes in mind. Appropriate instructional resources, teaching strategies and assessment procedures are suggested. All this is done considering the teaching-learning encounter as a sub- system of the general system of education which in itself comprises various areas of disciplinary persuasion.

All in all, attempts have been made in this unit to keep abreast with changing trends in teacher education. However, since the field of education is a dynamic enterprise, periodical review of the module is necessary to accommodate important emerging concerns. Welcome to the module.

ECT 201: CONTEMPORARY PEDAGOGIES IN TEACHER EDUCATION

WEEK	TOPIC
WEEK 0	INTRODUCTION - STUDENT CONTEXT & GOALS
WEEK 1:	DEFINITIONS OF CONCEPTS IN EDUCATION: GOALS, LEARNING OUTCOMES, TEACHING AND LEARNING
WEEK 2:	INSTRUCTIONAL MODEL: SYSTEMS APPROACH TO TEACHING AND LEARNING
WEEK 3:	INSTRUCTIONAL MODEL: SYSTEMS APPROACH TO TEACHING AND LEARNING
WEEK 4:	TEACHING STRATEGIES AND THEORIES - HEURISTIC AND EXPOSITORY STRATEGIES IN TEACHER EDUCATION.
WEEK 5:	INNOVATIVE LEARNING STRATEGIES FOR MODERN PEDAGOGY.
WEEK 6:	CHANGING TRENDS IN TEACHER EDUCATION
WEEK 7:	MOTIVATION AND SUSTAINING ATTENTION IN THE CLASSROOM
WEEK 8:	EFFECTIVE CLASSROOM COMMUNUICATION
WEEK 9	SYLLABUS INTERPRETATION AND PROFESSIONAL TOOLS
WEEK 10	INCLUSIVE EDUCATION – PROVIDING FOR INDIVIDUAL DIFFERENCES (Supporting ALL Learners)
WEEK 11	COMPETENCY BASED ASSESSMENT AND EVALUATION.
WEEK 12	REVISION
WEEK 13 & 14	EXAMINATIONS

OVERVIEW OF THE COURSE WEEK 0: INTRODUCTION - STUDENT CONTEXT & GOALS

This lesson is intended to introduce and orientate you to blended learning and to create a community of learners who will motivate each other during the course. You will be required to introduce yourself to your lecturer and colleagues either physically during face to face sessions or even online before other academic interactions start. This will be at the discretion of individual universities and lecturers. It will be important to also state your context and goals as well as what you think about language teaching and learning. You can also share any experience that you may have regarding the same.

WEEK 1: DEFINITIONS OF CONCEPTS IN EDUCATION: GOALS, LEARNING OUTCOMES, TEACHING AND LEARNING.

This lesson covers definitions and concepts like goals and learning outcomes, the relationship between teaching and learning and the nature and components of teaching and learning.

WEEK 2: INSTRUCTIONAL MODEL: SYSTEMS APPROACH TO TEACHING AND LEARNING

The second lesson looks at definitions of systems, examples, General Systems Theory, History of GST, Characteristics of a system, components of a system, impact to educational systems,. Further, we discuss systems approach to teaching/instruction, the school as a system and why teaching is considered as a system; planning to teach as a system.

WEEK 3: TEACHING STRATEGIES AND THEORIES - HEURISTIC AND EXPOSITORY STRATEGIES IN TEACHER EDUCATION.

This covers definitions of strategy – heuristic and expository; methods of teaching; experiential learning - Kolb's experiential learning.

WEEK 4: INSTRUCTIONAL METHODS: COMPETENCY BASED TEACHING METHODS.

This lesson addresses competency based teaching methods; enhanced teaching methods – what they are. Enhanced teaching methods such as problem solving, project based learning, innovative, collaborative and service learning.

WEEK 5: INNOVATIVE LEARNING STRATEGIES FOR MODERN PEDAGOGY.

It deals with innovative learning methods of teaching, modern pedagogy, crossover learning, learning through argumentation, incidental learning, context-based learning, computational learning. Learning by doing Science, adaptive learning, collaborative learning methods, cooperative learning, blended learning and e-learning.

WEEK 6: CHANGING TRENDS IN TEACHER EDUCATION

This lesson discusses key trends in teacher education, creating interactive learning environments for prospective teachers, development of face-face and hybrid pedagogies, using OERs in teaching and learning environments, planning and implementing large class pedagogy in inclusive classrooms.

WEEK 7: MOTIVATION AND SUSTAINING ATTENTION IN THE CLASSROOM

This lesson deals with the concept of motivation, the importance of motivation in teaching and learning, theories of motivation, motivation techniques in the classroom; sustaining attention.

WEEK 8: EFFECTIVE CLASSROOM COMMUNICATION

This lesson covers definitions, forms of communication, the process of communication, a model of communication, concepts of verbal and non-verbal communication, barriers to effective communication and effective communication principles.

WEEK 9: SYLLABUS INTERPRETATION AND PROFESSIONAL TOOLS

In this lesson, we guide the trainees on the syllabus, the scheme of work, the strand plan (formerly called Lesson plan), organization of teaching and records of work

WEEK 10: INCLUSIVE EDUCATION – PROVIDING FOR INDIVIDUAL DIFFERENCES (Supporting ALL Learners)

This lesson covers the concept of Inclusive Education; Types of individual differences among learners; Methods of identifying learners with individual differences; Methods of providing for individual differences.

WEEK 11: COMPETENCY BASED ASSESSMENT AND EVALUATION.

In this lesson, we look at the concept of competence based measurement, the formative evaluation and summative assessment procedures, competence based assessment and feedback

WEEK 12: REVISION

WEEK 13 & 14: EXAMINATION

These two weeks bring together the work you have been doing during the course to an end. This course unit will be examined and will partially contribute to the award of the degree of Bachelor of Education. It is anticipated that most Universities will have a minimum of 13 weeks' semester. We have therefore placed examinations in the last two weeks but Universities are allowed to go with their schedules. Kenyatta University examinations regulations will apply.

Troubleshooting

○ If you have problems accessing or logging into the system (LMS):

1. Go to the Student Help Desk -<https://support.ku.ac.ke/>
2. Chat with us at <http://chat.ku.ac.ke/>
3. Call the Digital school offices at (See our Contacts here) -<http://www.ku.ac.ke/dsvol/contact-us>
4. Guides about how to use moodle are available at <http://www.ku.ac.ke/dsvol/students-resources>
5. Also use our social media huddles:

Twitter: @KUDigitalSchool

Facebook: @KUDSVOL

MODULE LEARNING OUTCOMES

By the end of this module, you will be able to:

1. Apply technical concepts used in the process of teaching and learning.
2. Use systems thinking and approach in teaching and learning.
3. Use pertinent and contemporary appropriate approaches and strategies of teaching and learning in lesson delivery.
4. Prepare instruments for instruction
5. Motivate learners in the instructional process for problem solving
6. Select relevant contemporary instructional resources
7. Communicate effectively, creatively, and imaginatively in all instructional processes
8. Assess instructional needs and challenges of learners.
9. Develop effective strategies for classroom management and evaluation for addressing the needs of culturally diverse and special needs students.
10. Collaborate with peers in designing instruction and developing teaching

COURSE DESCRIPTION

This module contains content for Second Year B.Ed. students and 1st Year PGDE students. It will be offered during the 1st Semester. The major purpose of offering this course is lay the necessary teaching and learning among prospective teachers. Suggestions have been given on how to plan and implement any given educational curriculum with careful thought and foresight. This course responds to the Internetization of society and the penetration of digital technologies into learning as the most observable phenomenon in didactics and teaching methods. The focus is on Pedagogy of the twenty-first century that is sufficiently dynamic to respond to contemporary learners in tandem with changes in the development of national and world education. This course aims to provide student teachers with the foundations, principles, theories and general as well as innovative approaches to teaching and learning.

COURSE REQUIREMENTS

The general university entry requirements for B.Ed and PGDE apply. Further, as for any online course, you will be expected to have a tablet or computer to facilitate active participation in the course. The course is a *Blended Learning* one with both Face-to-Face and Online components. Forty percent of the entire semester learning will be online with the remaining 60% taking place on a face-to-face basis. In both phases, learners will be expected to construct knowledge from the links provided as well as other sources. Your lecturers will provide feedback during the teaching and learning encounters. Duration for learning tasks has been suggested in the module.

ASSESSMENT

It is important to note that the module has embedded learner formative assessment feedback tools that will enable you gauge your own learning progress. The tools include online collaborative discussions forums that focus on team learning and personal mastery and will therefore provide you with peer feedback, lecturer assessment and self- reflection. You will also be required to do one major assignment/project that is meant to assess the application of the skills and knowledge gained during the course. The project score in combination with scores for e-tivities (where graded) will account for 40% of your final examination score with the remaining 60% coming from a face to face sit-in final written examination that will be guided by your university examination policy and procedures. Assessment of learning progress will be undertaken from time to through quizzes and assignments with due notice being given. This is *formative assessment*. A *summative assessment* will be in form of a final university examination towards the end of the semester. We wish you the very best of experiences in this course.

LESSON 1

DEFINITIONS OF CONCEPTS IN EDUCATION: GOALS, LEARNING OUTCOMES, TEACHING AND LEARNING (Dr. Mary W. Nasibi)

1.1 Introduction

In this first lesson we will start by understanding the key concepts at the core of the course namely; goals, learning outcomes, teaching and learning. We will also look at the nature and components of teaching and learning and the relationship between the two processes.

1.2 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

- 1.2.1 Define goals, learning outcomes, teaching and learning
- 1.2.2 Explain the relationship between teaching and learning.
- 1.2.3 Explain the nature and components of teaching and learning

1.2.1 The definitions of goals and learning outcomes

Goals from education perspective can be defined broadly as statements of educational intension by a given education system. . They describe skills, attitudes and competencies students should attain during a given education program.

In Kenya, there are eight goals which educational programs from primary to university seek to achieve. They include but are not limited to the following;

- 1. Education should promote patriotism, nationalism and national unity.
- 2. Education should promote social, economic, technical and industrial needs for national development.
- 3. Education should promote individual development and self fulfillment
- 4. Education should promote sound moral and religious education.

However, in some countries goals can also be specific. In this case, they are formulated as statements with both an action verb and the competency required after a learning process. In such a case the goal takes the place of an objective e.g. by the end of the course the learner/ students should be able to explain/discuss ...

Learning outcomes refer to statements of what the learner should know and be able to do to demonstrate their knowledge, understanding, attitudes, values and skills and/or competences at the end of a learning process. Under the Competency Based Curriculum (CBC) in Kenya they have replaced objectives.

The learning outcomes like goals can be limited to a lesson, course, program or a degree or level of education. Therefore, they can be specific and narrow or general and broad. For example learning outcomes for early education according to Nyamoti (2018) include among the following:

- 1. Apply digital literacy for learning and enjoyment
- 2. Demonstrate basic literacy and numeracy skills in learning
- 3. Apply creative and critical thinking skills in problem solving

4. Practice hygiene, nutrition, sanitation, safety and nutrition to promote health and well being.

5. Practice appropriate etiquette for interpersonal relationships

Numbering, pacing and sequencing	1.2.1
Title	Definitions of goals and learning outcomes in education
Purpose	The purpose of this e-tivity is to enable you explain key concept in the course.
Brief summary of overall task	Read content on goals and learning outcomes under the links below. https://www.atikaschool.org/pre-primary-one-curriculum-with-syllabus/national-goals-of-education-in-kenya (NATIONAL GOALS OF EDUCATION IN KENYA ...) https://www.youtube.com/watch?v=g_Xm5IljYKQ&t=93s (Website) https://www.youtube.com/watch?v=2q-wPQhpIkQ&t=189s (Website)
Spark	
Individual task	a) Using bullet points, outline the key themes in the definitions of goals and learning outcome b) In a sentence develop your own definition of these terms: goals and learning outcome.

Interaction	a) Post two themes that are common in the definition of goals and learning outcomes. b) Provide positive and constructive feedback on the terms defined by stating the relationship between the two.
E-moderator interventions	1. Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas 3. Provide feedback on the learning progress 4. Close the e-tivity
Schedule and time	This task should take one hour.
Next	The concepts teaching and learning

It can also be specific to a lesson e.g. By the end of the lesson you will be able to describe ... E-

tivity 1.2.1 – Concepts of goals and learning outcomes

REGARDS

1.2.2 Concepts of teaching and learning

Teaching comes from the old English word ‘*tæcan*’ which means to show or point out. Learning on the other hand comes from the old English word *leornian* which means to get knowledge or to think about it. (Brown 2018)

Lefancons (1991) in Nasibi (2003) defines teaching as a process that facilitates changes in learners and entails telling and persuading, showing and demonstrating, guiding and directing learners’ efforts or a combination of these actions.

Learning on the other hand is defined by Ferrant (1980) in Nasibi (2003) as a process by which knowledge, understanding, attitudes, skills and capabilities can be acquired and retained..

Teaching and learning are therefore interconnected terms. One has to learn before teaching and therefore teaching is dependent on learning..

E-tivity 1.2.2

Numbering, pacing and sequencing	1.2.2
Title	Defining the term teaching and learning
Purpose	The purpose of this e-tivity is to enable you to explain the concept teaching and learning.

ODHIAMBO DARREN GREGORY

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Brief summary of overall task	Read more about the meaning of teaching and learning from the link below 1. https://www.researchgate.net/publication/272620585_by A.H Sequeira on introduction to concepts of teaching and learning (Website) 2. http://www.sun.ac.za/english/learning-teaching/ctl/t-l-(Website))resources/curriculum-t-l-assessment#:~:text=Teaching%20can%20be%20defined%20as,%2C%20delivery%2C%20assessment%20and%20reflection. b) Watch the video at the link attached: https://www.youtube.com/watch?v=-Xt4R2Eo0js
Spark	

Individual task	a) Using bullet points outline the key themes in the definitions of teaching and learning b) In a sentence develop your own definitions of the terms teaching and learning.
Interaction	a) Post two themes that are common in the definition of teaching and learning. b) Provide positive and constructive feedback on the definition of teaching and learning.
E-moderator interventions	1.Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas 3. Provide feedback on the learning progress 4. Close the e-tivity

Schedule and time	This task should take one hour.
Next	The relationship between teaching and learning

1.2.3 Relationship between teaching and learning

The relationship between teaching and learning is crucial. Both of them are processes, are interdependent and inseparable. Teaching and learning as a process starts with planning for teaching, the execution of the teaching and lastly carrying out assessment to find out if the teaching and learning outcomes have been realized.

Teaching and learning are interconnected and interdependent. They cannot operate independently. There can be no learning without teaching nor there teaching in the absence of learning. Both teachers and learners bring in knowledge of their learning experiences and interactions creating understanding and new experiences.

Although teaching determines the nature of learning and effective teaching leads on effective learning , a teacher cannot make learners to learn. The best he/she can do is to create an environment which will lead to learning; an atmosphere that will motivate students and contribute to active learning

According to Brown (2018), in teaching and learning relationship, the teacher has the role of a facilitator, human relation specialist and a motivator. The effective implementation of these roles will promote learning.

,E-tivity 1.2.3

Numbering, pacing and sequencing	1.2.3
Title	The relationship between teaching and learning
Purpose	The purpose of this e-tivity is to enable you to explain the relationship between teaching and learning.
Brief summary oral task	Watch videos1 and2 on these links. http://www.differencebetween.net/language/words-language/difference-between-teaching-and-learning/ (Website ; Brown 2018) https://onlinelibrary.wiley.com/doi/abs/10.1002/aehe.3640150403 (Website ; Brown 2018)
Individual task	a) Using bullet points, outline the key themes in the relationship between teaching and learning . b) In a sentence explain what you understand by each of the components.
Interaction	a) Post two themes that are common in the relationship of teaching and learning. b) Provide positive and constructive feedback on the explanation given on the relationship of teaching and learning.

E-moderator interventions	1.Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas 3. Provide feedback on the learning progress 4. Close the e-tivity
Schedule and time	This task should take one hour.
Next	The components of teaching and learning

1.2.3 The nature and components of teaching and learning

The nature of teaching and learning is not only complex, dynamic and flexible but also, interactive and creative. Although a teacher can be seen teaching , learning cannot be seen but can be inferred from change in observable behavior. It is difficult to understand how teachers teach and how their teaching is perceived by their learners. The nature of learning is characterized by the following: universality, experience, continuity, relative permanent change in behaviour and unobservable.

According to Nasibi (2003) the components of teaching and learning include stating the learning outcomes, identifying the content to be taught, selecting appropriate teaching methods and techniques, identifying instructional resources, presenting the lesson, measuring and evaluating the learning outcomes. **E-tivity 1.2.4**

Numbering, pacing and sequencing	1.2.4
Title	The nature and components of teaching and learning
Purpose	The purpose of this e-tivity is to enable you to describe the nature and explain the components of teaching and learning.
Brief summary orall task	Watch videos1 and2 on these links. <u>http://www.differencebetween.net/language/words-language/difference-between-teaching-and-learning/(Brown 2018)</u> <u>https://onlinelibrary.wiley.com/doi/abs/10.1002/aehe.3640150403</u> (Website)
Spark	

Individual task	a) Using bullet points, outline the key themes in the nature and components of teaching and learning b) In a sentence explain what you understand by each of the components.
Interaction	a) Post two themes that are common in the nature and components of teaching and learning. b) Provide positive and constructive feedback on the explanation given on the components of teaching
E-moderator interventions	1, Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas 3. Provide feedback on the learning progress 4. Close the e-tivity
Schedule and time	This task should take one hour.
Next	The next lesson will be on GANAG, Instructional Model; Systems Approach to teaching

Assessment Questions

1. Differentiate between a goal and a learning outcome
2. State the other four goals of education apart from the four given in the notes, 3. Highlight key features in the relationship between teaching and learning.
4. Explain the following components of teaching:
 - a) The content
 - b) The methods of instructions

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LESSON 2

INSTRUCTIONAL MODEL: SYSTEMS APPROACH TO TEACHING AND LEARNING By Dr. Adelheid M. Bwire

2.1 Introduction

Systems thinking involves moving from observing events or data, to identifying patterns of behavior overtime, to surfacing the underlying structures that drive those events and patterns. By understanding and changing structures, we can create satisfying, long-term solutions to chronic problems in education. Systems thinking perspective requires curiosity, clarity, compassion, choice, and courage. This approach includes willingness to see a situation more fully, to recognize that we are interrelated, to acknowledge that there are often multiple interventions to a problem in education, and to champion interventions. This lesson explores how to look at education and classroom teaching as a system

2.2. Expected learning outcomes

By the end of the lesson, you should be able to:

- 2.2.1 Define a variety of terms in systems approach
- 2.2.2. Discuss various components of a system in education
- 2.2.3. Apply systems thinking in instructional planning

2.2.1. Definitions:

System

Romszowski (1981): “a set of elements or components or elements or objects which are interrelated and work towards an overall objective”

Groenewegen (1993): “a complex of factors interacting according to an overall plan for a common purpose or goal”

“an organized set of doctrines, ideas, or principles usually intended to explain the arrangement or working of a systematic whole” (Merriam-Webster Dictionary)

Generally: Any collection of **interrelated parts** that together constitute a larger **whole**. These component parts, or **elements** of the system are intimately linked with one another, either directly or indirectly, and any **change in one** or more elements may **affect** the overall performance of the system, either beneficially or adversely. The systematic approach to teaching provides a method for the **functional organization and development of instruction**.

Examples of systems

Common systems used in our day to day activities are for example:

The human body – and subsystems such as the circulatory system, digestive system and nervous system
Mechanical systems: refrigeration system, generators, music system, computer system, engines etc.
Social systems: organisation such as family, education systems, political parties, trade union movements, industries and churches

Natural systems: rivers, forests etc.

General Systems Theory

General systems theory (GST) was outlined by Ludwig von Bertalanffy (1968). Its premise is that complex systems share organizing principles which can be discovered and modeled mathematically. To quote Bertalanffy, “...there exist models, principles, and laws that apply to generalized systems or their subclasses, irrespective of their particular kind, the nature of their component elements, and the relations or “forces” between them.” (Bertalanffy, 1968, pp 32).

ODHIAMBO DARREN GREGORY

+254704597085

Systems theory is the interdisciplinary study of systems. A system is a cohesive conglomeration of interrelated and interdependent parts which can be natural or human-made. Every system is bounded by space and time, influenced by its environment, defined by its structure and purpose, and expressed through its functioning. Changing one part of a system may affect other parts or the whole system. The goals of systems theory are to model a system's dynamics, constraints, conditions, and to elucidate principles (such as purpose, measure, methods, tools) that can be discerned and applied to other systems at every level of nesting, and in a wide range of fields for achieving optimized **equifinality**.

History of General Systems Theory

Systems Theory comes from the General Systems Theory proposed by the biologist Ludwig von Bertalanffy (1968).

There is a need for a **unified** and **disciplined inquiry** in understanding and dealing with increasing complexities in organisations.

The systems view investigates the **interaction between components of an organisation, and the relation of components to their larger environment**.

3.2. Characteristics of general systems theory

- Open system: a system keeps evolving and its properties keep emerging through its interaction with environment
- Holistic view: systems theory focuses on the arrangement of and relations between the parts that connect them into a whole. The mutual interaction of the parts makes the **whole bigger than the parts** themselves.
- Goal-directedness: systems are goal oriented and engage in feedback with the environment in order to meet the goals. Also, every part of the system is interdependent with each other working together toward the goals.
- Self-organizing: productive dynamic systems are self-organizing this implies the adaptive ability of the systems to the changes in the environment.

2.2.2. Components of a system

1. **Goals** and mission: every system has a goal (target to be achieved). All members work towards the achievement of the set goal.
2. **Elements**: there are several elements – these are interrelated and interact. Exists in a hierarchy of relationships – each functional unit forms a subsystem with elements that cannot work independently. A supra-system: one that has subsystems each of which should perform but is interdependent on other subsystems.
3. Has boundaries – which distinguish it from other systems parts ensuring that a specific task is performed at a particular point, but both/all parts depend on each other
4. Environment: a set of conditions such as resources, changes and constraints (e.g. inadequate material, dissenting voices) that affect the system either positively or negatively. There is a flow both inward and outward the system into the environment. In order to survive, the system must interact with and adjust its environment and the other parts of the supra system. The environment provides inputs such as money, people and resources which will determine the quality of the operation of the system

5. **Harmony:** a coherent interaction for attaining the common goal. Elements work in harmony, although each has its own function which it contributes in achieving the goal of the system.
6. Some degree of disorder (**internal entropy**) – because systems are open and operate within environmental constraints; there is a critical point at which a system may collapse (hence need for flexibility of objectives of a system to minimise entropy)
7. **Feedback** – input or information about the achievement (output) for purposes of re-examining the system. Provides for assessing the suitability of the success level in the interaction leading to the attainment of the intended goal.
8. Growth- through either transformation or diversification or multiplication
9. Dynamic stability (equilibrium) with the environment – in harmony with the environment and has to exhibit levels of balance among its elements/subsystems
10. It has a set of inputs (learners) which are subject to a process in order to attain certain objectives which appear as outputs (educated students).
11. It has a self-adjusting combination of interacting people/institutions and things to accomplish a pre-determined purpose. (Hooper (1971))

2.2.3. Impact to Educational Systems: Systems Approach to Teaching/Instruction According to Banathy (1987), there are four subsystems in any educational enterprise:

1. The **learning experience subsystem:** the cognitive information processing of the learner
2. The **instructional subsystem:** the production of the environment or opportunities for learners to learn by the instructional designers and teachers
3. The **administrative subsystem:** decision making of resource allocation by the administrators based on the instructional needs and governance input
4. The **governance subsystem:** the production of policies which provide directions and resources for the educational enterprise in order to meet their needs by "owners"
5. The instructional system is part of educational system. Reigeluth (1996)

Inputs to the educational systems approach include:

- Well defined objectives, ○ Analysis of the intended audience,
- Special criteria desired by the customer (government, parents, students), ○ Analysis and use of existing resources - Content requirements, Content development and Use ○ A team of instructional system specialists, subject matter experts, writers, and SNE specialists

The School as a system

- It is an organised entity with goals received from society (MoE – see national goals, curriculum and syllabuses); receives raw materials (learners) from the environment; learners are put through a process; the output is assessed to see if behaviour is transformed.
- It has elements/parts which must work together for the goals to be realised (head teachers, teachers, support staff, learners, T/L resources, physical resources etc). All these must interact in order to realise the desired output.
- It must have harmony – all elements (all those involved in the education system) must work harmoniously for goals to be realised. E.g. Head teacher as chief administrator must provide

ODHIAMBO DARREN GREGORY

+254704597085

leadership etc. Learners must work alongside teachers etc. Support staff must perform their respective roles etc.

- It must provide feedback – both teachers & learners must get feedback to check if the products are being processed desirably. Feedback maybe positive or negative. Then teacher can adjust teaching methods, resources, experiences, classroom management to achieve desired goals.

Why Teaching is considered a system

- Because it has a well-defined goal or objective;
- It consists of more than one element
- All these elements work in harmony.
- It also has provision for feedback.

NB: There is an interrelationship among all the components. The focus of the systematic instructional planning is the STUDENT/LEARNER.

Parts of a systematic instruction explained:

The following procedure guides the teacher in accomplishing instructional goals:

1. Specification of objectives: Define Objectives - Instruction begins with the definition of Instructional Objectives that consider the learners' needs, interests and readiness.
2. Selection of content: according to the curriculum
3. Assessment of entry behaviour: organizing students into groups.
4. Choice of teaching strategy - On the basis of the objectives, the teacher selects the appropriate teaching methods to be used. And, in turn, based on the teaching method selected, the appropriate learning experiences and appropriate materials, equipment and facilities (resources) will also be selected. The strategic method or learning methodology is based on the nature of the learner. Chose appropriate experiences; allocation of time and learning space.
 - Assign personal roles – The use of learning materials, equipment and facilities necessitates assigning the appropriate personnel to assist the teacher and defining the role of any personnel involved in the preparation, setting and returning of these learning resources.
5. Implement the instruction - Carrying out or Executing teaching instruction. With the instructional objectives in mind, the teacher implements planned instructions with the use of the selective teaching method, learning activities, and learning materials with the help of other personnel whose role has been defined by the teacher.
 - *Examples of Learning Activities:* Reading, Writing, Interviewing, Reporting or doing Presentation, Discussing, Thinking, Reflecting, Dramatizing, Visualizing, Creating, Judging, evaluating
 - *Examples of Learning Resources for Instructional Use:* Textbooks, Workbooks, Programmed materials, Computer, Television programmes, pictures, Slides and transparencies, Maps, Charts, Cartoons, Posters, Models, Mock ups, Flannel board materials, Chalkboard, Reals objects
6. Evaluate outcomes/Evaluation of teacher and learner performance (quality control monitoring, performance testing etc.) - after instruction, teacher evaluates the outcome of instruction. From the evaluation results, teacher comes to know if the instructional objective was attained.

7. Analysis of feedback by teacher and learner: refining the process - It is dependent on the outcome. If the instructional objective was attained, teacher proceeds to the next lesson going through the same cycle once more. If instructional objective(s) was not attained, then teacher diagnoses what was not learned and finds out why it was not learned in order to introduce a remedial measure for improved student performance and attainment of instructional objectives. The teacher should device plans (remedial, tutorial or give more examples) for the learners to fully understand the lesson. (Glaser, 1962)

NB: The purpose of a system instructional design is “to ensure orderly relationships and interactions of human, technical, and environmental resources to fulfill the intended goals which have been established for instruction.” (Brown, 1969)

The phases or elements are connected to one another. If one element or one phase of the instructional process fails, the outcome, which is learning, is adversely affected. The attainment of the learning objective is dependent on the synergy of all the elements and of all the factors involved in the process

Planning to teach as an example of part of the instructional system The lesson plan

The teacher/instructor should plan to teach through the scheme of work (long-term plan for content activities and resources in teaching e.g. in one term or year).

For immediate classroom use, the teacher/instructor prepares a lesson plan. This is a plan indicating the instructional objectives, time required, the major steps to follow, the content and relevant activities, the necessary resources and teaching aids. The plan also gives room for self-evaluation.

Every lesson plan or lesson presentation must have objectives, elements, harmony (suitable methods, resources, entry behaviour have to be in harmony), evaluation. **Lesson presentation as a model systems approach**

A well-presented and organised lesson should exhibit the following qualities:

- Relevance to objectives.
- Accuracy of information/content.
- Likelihood to arouse and maintain interest. o Learner participation in: Oral activities, Written experiences, Listening activities, Visual activities, Art and craft activities e.t.c.
- Appropriate pace for learners.
- Clarity of organization of information - clarity of expression, conversational, of quality.

Models

Figure 1: A simple Model of a systems approach



In Figure 2 below, the system consists of four distinct elements A, B, C, D, which are related to or dependent upon each other as indicated. Note that some interrelationships may be two-way, while others may be one-way only. These elements may themselves be capable of further breakdown into other smaller components, and may thus be regarded as **sub-systems** of the overall system.

The processes of teaching and learning can be considered to be very complex systems indeed. The **input** to a given teaching/learning system consists of people, resources and information, and the **output** consists of people whose performance or ideas have (it is to be hoped) improved in some desired way. A schematic representation of systems of this type is shown in Figure 3:

A simple system is illustrated schematically in Figure 2:

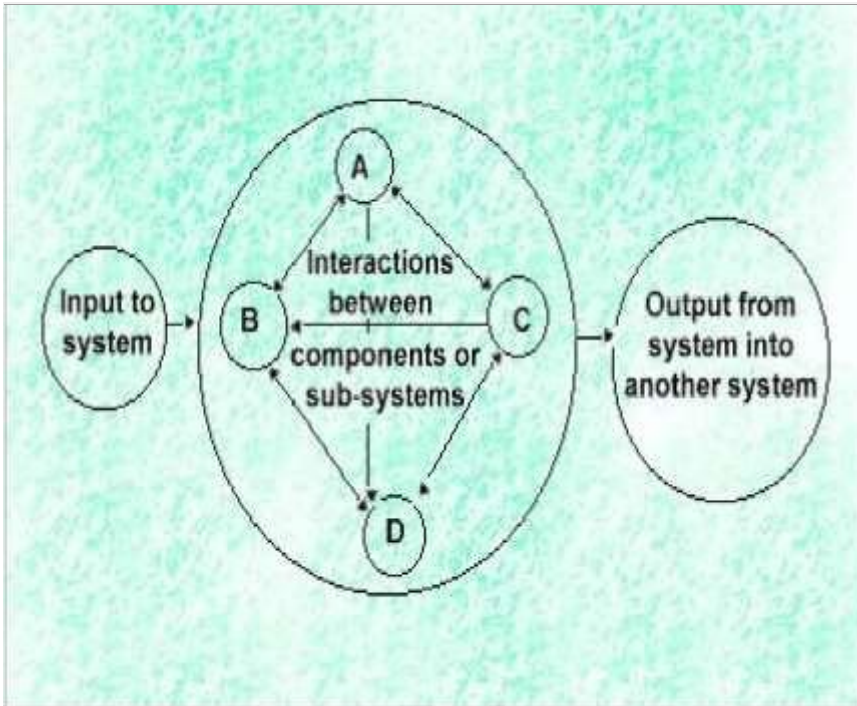


Figure 2. A typical system

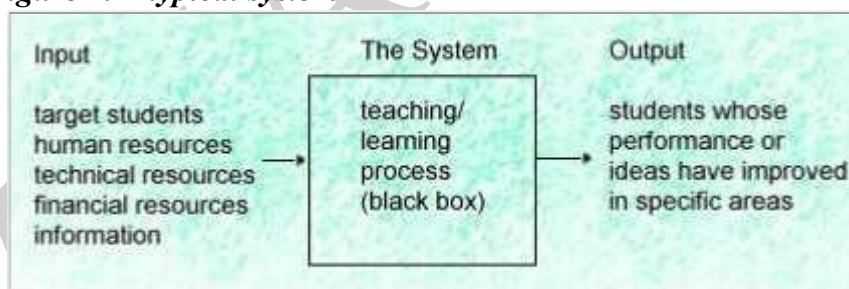


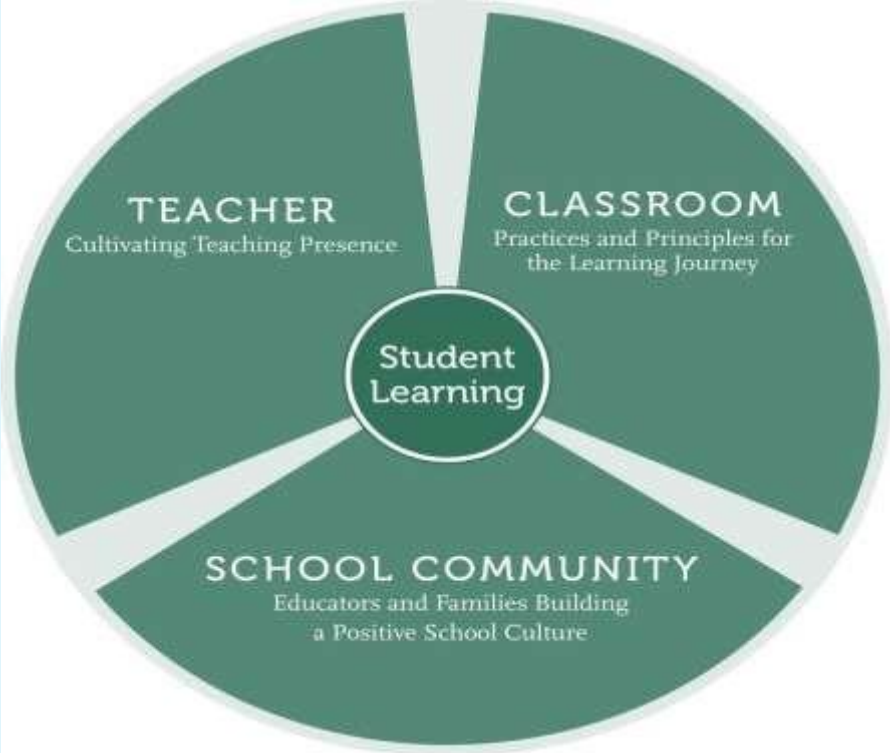
Figure 3. The 'systems' model of the educational process
(Romiszowski 'Designing Instructional Systems')

E-tivity: 2.2.1: INSTRUCTIONAL MODEL: SYSTEMS

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APPROACH TO TEACHING

Numbering, pacing and sequencing	2.1.2: The nature and components of teaching and learning
Title	The nature and components of teaching and learning
Purpose	The purpose of this e-tivity is to enable you to describe the nature and explain the components of teaching and learning.
Brief summary oral task	Follow the link below for more insights. https://eric.ed.gov/?q=systems+approach+to+teaching&ft=on&id=ED124184 and https://eric.ed.gov/?q=systems+approach+to+teaching&pr=on&pg=2&id=EJ1161225
Spark	
Individual task	a) Using bullet points, outline the key themes in the nature and components of teaching and learning b) In a paragraph explain how a successful system works.
Interaction	a) Post two themes that are common in the nature and components of a system b) Provide positive and constructive feedback on the explanation given on the components of a system
E-moderator interventions	1, Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas

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	3. Provide feedback on the learning progress 4. Close the e-tivity
Schedule and time	This task should take one hour.
Next	The next lesson will be on TEACHING STRATEGIES AND THEORIES-

Assessment Questions:

1. *What are the steps of the instructional process or the parts of a systematic instruction?*
2. *What does each step mean?*
3. *Are the elements or phases of instruction in the systems approach design independent of one another or do they relate to one another?*

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- (3) <http://www2.rgu.ac.uk/celt/pgcerttlt/systems/sys3.htm>
- (4) <http://edutech202.blogspot.in/2012/12/the-concept-of-system-approach-to.html>
- (5) <http://mjthepedagogue.blogspot.in/2011/02/approaches-of-educational-technology.html>
- (6) <http://www.slideshare.net/maheshnangude/types-of-educational-technology>
- (7) <http://www.slideshare.net/FaKhalid/approaches-in-educational-technology>

LESSON 3

TEACHING STRATEGIES AND THEORIES - HEURISTIC AND EXPOSITORY STRATEGIES IN TEACHER EDUCATION. (Dr Harrison M.Mulwa)

3.1. Introduction

Most learning and teaching strategies can be broadly classified under two categories. These are: Heuristic Strategy and Expository Strategy. Many teaching methods fall under the two strategies. Some of the teaching methods are a blend of the two strategies.

3.2 Learning Outcomes

By the end of this lesson you will be able to:

- 3.2.1 Identify Heuristic Methods of teaching
- 3.2.2 Identify Expository Methods of teaching
- 3.2.3 Apply Kolb's Theory of Experiential learning to teaching
- 3.2.4 Explain the conveniences and inconveniences of Heuristic Methods
- 3.2.5 Explain the advantages and disadvantages of Expository Methods

3.2.1. Heuristic Methods of teaching

Definition Strategy in teaching

A strategy in teaching can be defined as the general idea, or the overall way in which the process of teaching and learning is organized. In teaching, one finds some common aspects of practices across a variety of methods. One can therefore say that a strategy encompasses a number of methods which share common features, even though the features may apply differently, in each method.

Definition of Method in teaching

A method can be described as a way, which, practical in many instances, a teacher follows in the delivery process of content in a given strand or lesson. It is seen as a means to achieving the learning outcome, in specific steps, deliberately arranged to lead to a desired aim, goal or end.

Heuristic Strategy

The heuristic strategy, commonly called "Discovery Strategy" refers to teaching approaches or methods which encourage the teacher to allow learners to discover facts and information, in the process of instruction. In this strategy, the learner and not the teacher, is the centre of instruction. The term "heuristic" has been coined from the Greek word <eureka> meaning to find out or to discover. The teacher then becomes a facilitator, who through the skills acquired in his/her training, guides the learners to find out for themselves the answers to issues, rather than telling them directly.

3.2.1: E-tivity: Heuristic Strategy

Title	Identifying heuristic methods.
Purpose	To enable the learner to apply heuristic methods
Brief Summary of Overall Task	To establish the overall nature of heuristic methods
Spark	<i>Know when to be silent</i>

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Individual Task	Read the following reference and answer the questions. 1.Choose and study any 4 of the methods. List 3 characteristics of each , which show they are Heuristic in nature. https://wabisabilearning.com/blogs/literacy-numeracy/28-instruction-methods-infographic
Interaction Begins	Discuss the question with your peers. Post your answers on the <i>Kusoma forum</i> before the end of the week of this lecture.(<i>kindly avoid any unrelated chat here</i>)
E-Moderator Interventions	Your opinions will be summarized here after you've posted them
Schedule and Time	This should take 2 hours
Next	Read also <i>Instructional Methods in Education</i>. Chapter 4 for further information(<i>this text is available in the library, consult your lecturer</i>)

3.2.1 E-tivity Heuristic Strategy

Title	Advantages/disadvantages of Heuristic methods
Purpose	To enable learner to determine when to use and when not to use heuristic
Brief Summary of overall Task	Study the conveniences/inconveniences
Spark	<i>Make value judgements</i>
Individual Task	Read the article and answer the following questions:- https://www.preservearticles.com/education/advantages-and-disadvantages-of-heuristic-method-of-teaching-science/5381 1. How do these advantages apply in your own subject area? 2. How would the disadvantages affect teaching of your subject?
Interaction begins	Discuss the questions with peers and share your opinions on the forum
E-Moderator Interventions	To view and summarise answers on forum.
Schedule	This task should take 3 hours.
Next	Ask your lecturer to guide you on access to the text <i>Instructional methods in Education. Read chapter 4 for further information</i>

3.2.2 Expository Strategy.

The Expository Strategy involves direct teaching. The teacher does most of the talking. In the meantime, learners remain largely inactive. In this Strategy learners have to memorize information with the aim of reproducing the same, when called upon to provide explanations. As expected, learning with this strategy will most likely produce a student who has weaknesses in making value judgments on what they have learnt, among others. This strategy leads to shallow learning since it does not lead learners to express their views and opinions on what they are told. The Expository Strategy can be said to be teacher –centered in nature.

3.2.2. E-tivity . Expository Strategy

Title	Identifying Expository methods.
Purpose	You will be able to apply expository methods
Brief Summary of overall Task	To establish the overall nature of expository methods
Spark	<i>Who wants to be a passive listener?</i>

Individual Task	Read the and answer the questions. http://www.nzdl.org/gsdmod?e=d-00000-00--off-0fnl2%2E2--00-0---0-10-0---0---0direct-10---4-----0-11--11-en-50 1. From the descriptions which are Expository methods? 2. Give reasons for your choice
Interaction Begins	After exchanging your views on the questions with peers post your answers, in short sentences on the forum.
E-Moderator Interventions	Your answers will be summarized in the forum by your e-moderator.
Schedule	The activity should last 2 hours
Next	Read <i>Chapter 4 of Instructional Methods in Education</i> for further information.

5.6.2. E-tivity Expository Strategy

Title	Advantages/Disadvantages of Expository
Purpose	To allow learner to determine when to and not use Expository methods
Brief Summary of Overall Task	To be able to determine when to use or not use Expository methods
Spark	<i>Consider what is relevant to the situation</i>
Individual Task	http://www.nzdl.org/gsdmod?e=d-00000-00---off-0fnl2%2E2--00-0---0-10-0---0---0direct-10---4-----0-11--11-en-50 1. Study the methods and find out which expository methods would be advantageous to use in teaching one of your subjects. (Note that not all the methods here are expository) 2. Give reasons why you chose this/these methods
Interaction	Discuss the questions with your peers
E-Moderator Intervention	To summarise student findings and post them on forum by end of lesson week.
Schedule	Post your group answers in the forum. This task should last 3 hours
Next	Read also <i>Instructional Methods in Education Chapter 4</i> for additional information.(find out from your lecturer how to access this text)

3.2.3. Experiential Learning

Definition of Experiential Learning.

Experiential Learning can be defined simply as <Learning through experience> or <learning through reflection on doing>. Key words here are **reflection** and **doing**. The learner then is not just simply doing something but up and above that he/she is reflecting or thinking, evaluating, judging, at every step of the way in the learning process

Kolb's Theory of Experiential Learning

A number of theories have been developed over the last few decades to support this Learning method. One of the most applied theories applied with this method is **David Kolb's Theory of Experiential Learning**. It is one of the best known educational theories in education. The theory presents a way of structuring a session or a whole course using a learning cycle. The different stages of the cycle are associated with distinct learning styles. Individuals differ in their preferred learning styles, and recognizing this is the first stage to developing a strand that will cater for the different learning styles represented among your learners. . In this Lesson we will attempt to see how this theory can be applied to teaching and learning to improve learning .

The Experiential Learning Cycle

Kolb's experiential learning style theory is typically represented by a four-stage learning cycle in which the learner comes into contact with all the contents of the programme. These stages are as outlined below.

1. **Concrete Experience** - a new experience or situation is encountered, or a reinterpretation of existing experience.
2. **Reflective Observation of the New Experience** - of particular importance are any inconsistencies between experience and understanding.
3. **Abstract Conceptualization** reflection gives rise to a new idea, or a modification of an existing abstract concept (the person has learned from their experience).
4. **Active Experimentation** - the learner applies their idea(s) to the world around them to see what happens.

3.2.3 E-tivity ;Kolb's Experiential Learning Theory

Title	Kolb's Theory
Purpose	You will be able to apply this theory to teaching and learning
Brief Summary of overall Task	This task allows you to understand the components of experiential learning and its application to teaching
Spark	<i>Every great success begins with a mental image picture of that success.</i>

Individual Task	Read the <u>article</u> . Answer the questions below 1. Choose an area from one of your subjects. 2. How can you apply the four stages mentioned to improve learning in that particular area?
Interaction Begins	Discuss the questions with members of your peer group and post your answers in the forum.
E-Moderator Interventions	Your opinions will be summarized and posted on the link you'll have been given
Schedule	This task will last 3 hours.
Next	This <u>link</u> is also crucial to understanding the nature of Experiential learning.

Quiz

1) What can you do to ensure success of a laboratory experiment?

- preview experiment
- give instructions on how to proceed
- assess results after experiment

2) Heuristic methods are useful since they involve the learner. Discuss.

-There is deep learning here. The learner is answering on the why and wherefore of facts.learner has a chance to think, reflect on ideas thus being in charge of his own learning.

3) The lecture method is best for covering a lot of content in any given field. What are possible drawbacks to this assertion?

- The learner is a passive listener
- The learner has little time to reflect and digest on what he is learning

4) 'When I do a demo in class, I am assured that learners have understood everything'

Comment on this teacher's report.

-The learners may not have understood all since they did not do the action themselves. They needed to be allowed to have hands -on experience for the teacher to be fully sure learning has taken place.

5. Think of 2 areas of any of your teaching subjects and say how you can apply experiential teaching to improve learning.

Consider how you would guide your learners through these steps in order to concretise the learning;

ODHIAMBO DARREN GREGORY

+254704597085

Experience----reflection on the experience-----Conceptualisation.....Experiment to prove the first three steps.

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LESSON 4

INSTRUCTIONAL METHODS: COMPETENCY BASED TEACHING METHODS.

By Dr. D. Muindi

4.1 Introduction

In chapter four of this module, we learnt about teaching strategies and teaching methods. We defined heuristic and the expository strategies of teaching and explained that heuristic strategy is a learner centred approach to teaching whereas the expository strategy is a teacher centred approach to teaching. Under each strategy, we discussed several teaching-learning methods which you can use to present information to the students. We concluded that no one teaching-learning methods is suitable for every teaching situation.

In this topic, we shall explore more instructional methods that lean towards competence-based teaching-learning methods. In particular, we shall learn about enhanced teaching methods such as problem solving, innovative, collaborative and cooperative learning methods, project-based learning and service learning. However, before we delve much in the methods, let's first explore the concept of competence-based teaching methods.

4.2 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

- 4.2.1 Explain competence-based approach to learning and give its benefits.
- 4.2.2 Identify the enhanced teaching methods and explain how they enhance learning.
- 4.2.3 Explain each of the six given innovative methods and give their benefits.
- 4.2.4 For any four of the six given learning enhancement methods give the steps involved in using them.

4.2.1 Competence Based Teaching Methods.

In our earlier topics, we learnt that a teaching method comprises of the principles and methods used by a teacher to enable the students acquire knowledge and skills. A teaching-learning method is determined by the complexity of the subject matter, the instructional media available as well as the learning needs of the learners and the objectives to be achieved.

Competency based educational (CBE) approach allows learners to advance at their own pace regardless of the learning environment, based on their ability to master the skills. This approach is tailored to meet learning abilities and can lead to more efficient student outcomes. CBE recognizes that learning can happen in many different ways and rewards demonstration of competency rather than the time spent learning.

What Is Competence Based Education?

Competence based approach in education refers to teaching and learning of measurable skills rather than abstract learning. The task to be completed must create the outcome that can be identified and measured. The learning task needs to be meaningful and engaging and to be completed in an effective and efficient manner.

Competency based learning has been described as very flexible since it depends on the individual learner. It is skill based and the focus is not the final outcome but the journey. In addition, the student is more engaged in the learning process besides

4.2.2 Enhanced Teaching Methods

The following six learning methods are given as good examples of competency-based learning methods. i Problem solving learning. ii Innovative learning iii Collaborative learning iv Cooperative learning (case studies) v Project based learning method vi Service based learning method

Let us now look at each of these learning methods as we explore more on the competencebased learning.

Problem Solving Learning.

Problem solving learning is defined as the act of defining a problem, determining the cause of the problem, identifying, prioritizing and selecting alternatives for the solution and implementing the solution. The student is engaged in the entire journey from defining the problem up to the implementing the solution. A four-step problem solving process and methodology is often cited as very effective in this learning method

Problem Solving Learning Method

Problem solving approach is a student centred learning strategy which requires the student to become an active participant in the learning process. The student is presented with a problem which requires him/her to find the solutions (searching for information).

The student observes, understands, analyzes, interprets and finds solutions and performs applications that lead to holistic understanding of the concept and in the process, new knowledge is learnt. Through the method, scientific process skills are developed.

5.5.1.2 Steps Involved In The Problem-Solving Learning Method.

The process of problem solving involves five basic steps. i

Assessment

ii Problem identification, iii

Planning iv Implementation

and v Evaluation.

Benefits Of The Problem-Solving Learning Method.

Problem solving skills are among the most valued skills because they can be applied in various real life situations. Life is about solving problems. Learning about problem solving approaches therefore would equip you with tools to help you strategize solutions when in difficult problem circumstances.

Project Based Learning Method

The project method of learning is one which enables students to acquire and apply knowledge and skills to define and solve realistic problem using a process of extended inquiry. In the method the

teacher encourages the student to apply their growing skills in purposeful ways. It plays an important role at the initiation, implementation and culmination parts of the project.

Benefits Of The Project Method

The primary aim of the project learning method is to make students think independently as they use acquired knowledge and skills to solve problems. As a learning method, the project method has several benefits.

4.2.3 Innovative Learning Methods

An innovative learning method is about solving given problems in a different way. This calls for creativity in the problem-solving skills. Innovation therefore involves a different way of looking at a problem and solving it.

Collaborative Learning Method

This is a learning method that groups students to work together towards a common goal. The learning method is based on the collaborative learning theory which emphasizes that: a. knowledge is a social construct.

- b. Educational experiences involve interaction and social exchange.
- c. Contextually relevant and engaging learning activities are student centred and lead to deeper learning.

In addition, experts claim that:

- i the active exchange of ideas within groups of students promotes critical thinking.
- ii Teams that engage in cooperative learning achieve; ○ Higher levels of thought and ○ Retain information longer than students who work solely as individuals.

Strategies For Effective Collaborative Learning

Collaborative learning is hard work which does not just happen. It has to be intentionally designed so as to be effective. There are strategies to promote effective collaborative learning.

Advantages Of Collaborative Learning.

This method helps students learn how to work effectively as a team. Cooperative learning is a specific kind of collaborative learning where students work together in small groups on a structured activity.

The students are individually accountable for their work and the work of the group as a whole is also assessed. Cooperative groups work face to face and learn to work as a team.

Service Learning.

Service learning actively involves students in a wide range of experiences, which benefits others and the community while advancing the goals of a given curriculum.

It offers direct application of theoretical models.

Community based service activity are paired with structured preparation and students reflection.

Proponents of academic service learning feel that:

- i Real world application of classroom knowledge in a community setting allows students to synthesize course materials in more meaningful ways.
- ii Common goals achieved by service learning are;
 - Gaining deeper understanding of the course/curricular content.
 - A broader appreciation of the discipline.
 - An enhanced sense of civic responsibility.

Qualities Of Service Learning

Several qualities of service learning have been given as follows:

- i Reflective ii Integrative. iii Contextualized vi Strength based v Reciprocal vi Life long

Some examples of service learning are as follows:

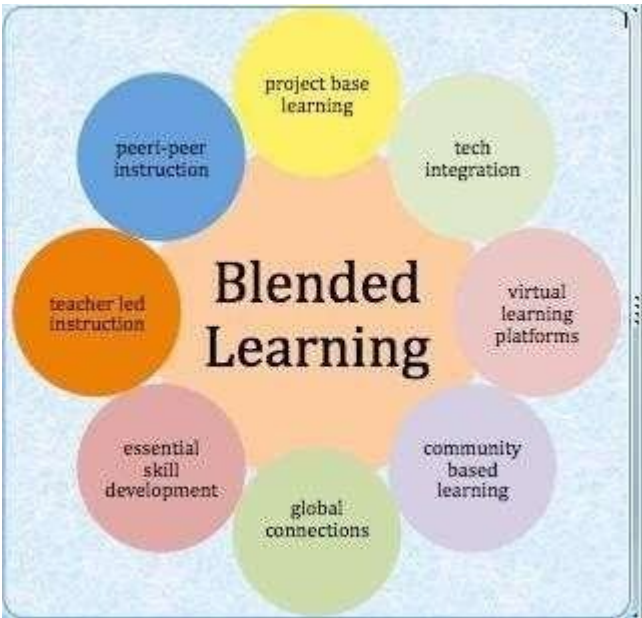
- i Field trip ii Research project
- iii Events iv Crafts v Fundraisers vi Awareness campaigns
- vii Various service projects such as cleaning, building and gardening

Benefits Of Service Learning

The following are some of the benefits of service learning.

- i Enables development of skills in critical thinking, problem solving, leadership, decision making collaboration and communication.
- ii Building of relationships with community members. iii Connecting experiences to academic subjects. vi Developing deeper understanding of themselves and empathy and respect for others 4.4.2

Numbering, pacing & sequencing	4.4.2 Competency based teaching and learning
Title	Instructional Methods.
Purpose	The purpose of this e-tivity is to understand the importance of competency-based learning and be able to choose appropriate teaching competency based method for effective learning by the student based on the learners learning needs, content complexity and instructional media available.
Brief summary of the overall task	Read the provided links https://www.waldenu.edu , www.tonnybates.ca , https://asq.org , https://www.slideshow.net , https://online.nwmissouri.edu , www.intechopen.com https://impact.cjv.muni.cz and 1. Compare the learners' roles between competency based
	learning methods and traditional learning approach. 2. For each of the six learning methods discussed in this Lesson and i. For each identify the benefits and challenges of the learning method. ii. Explain the steps involved in using each learning method.
Spark	

	
Individual task	□ Using the content in one of your teaching subjects, identify two topics, where one can be learned through the collaborative and the other cooperative learning methods. In the two learning situations, identify the key differences that would be observed between the two learning methods.
Interaction begins	Provide your assessment on the suitability of the topic given by the team learners in which experimentation would be done through remote lab method. Do this on the discussion forum 1.3 on the Kusoma platform
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress. 4. Close the e-tivity.
Schedule and time	This task should take two hours.
Next	

Questions

1. Compare teachers and the students' roles in traditional learning approaches and the competence based teaching methods
2. Prepare a flow chart to show the steps involved in problem solving method
3. Identify all the steps that you must go through in order to solve the provided problems.
4. Explain the differences between the collaborative and the cooperative learning methods.
5. Discuss the benefits of the service learning to the students.

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ODHIAMBO DARREN GREGORY

+254704597085

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LESSON 5

INNOVATIVE LEARNING STRATEGIES FOR MODERN PEDAGOGY.

By Dr. D. Muindi

5.1 Introduction

The 21st century is characterized by great strides in the fields of technology and scientific discoveries, globalization, internetization, informatization and explosion of knowledge and artificial intelligence. In particular, the computers, smart phones, video games, internet search engines have all redefined how we access information. The emerging digital technologies environment has a major impact on how learning in the classroom takes place as visual perception and discussion of knowledge replaces the traditional reading of text books have affected every sphere of life including in education and created immense pressure on how the teacher – learner interaction in the classroom takes place. In addition, the 21st century world requires innovativeness and creativity for the individuals and nations to survive in a fast changing global environment. In this global village, Kenya cannot therefore be left holding onto dysfunctional ways which have failed in enabling development in science and technology.

As the students gets exposed to alternative and interactive sources of information, it is not possible for the traditional teaching approaches which heavily rely on promoting rote (memory) learning to remain intact. The teacher has to therefore adapt innovative teaching approaches that are greatly influenced by the changed digital environment. The characteristics of such modern teaching methods are: i Learner centred ii Task based or activity based iii Resource based. iv Interactive in nature v Integrative in nature vi Involve peer collaboration

5.1 Lesson Learning Outcomes

By the end of this lesson, you will be able to:

- 5.2.1 Explain the meaning of the concepts ‘innovative learning strategies’ and ‘modern pedagogy’.
- 5.2.2 Give the characteristics of innovate learning methods and show that each of the eleven provided learning methods as presented under 8.4 to 8.12 qualifies as an innovative learning method.
- 5.2.3 Explain each of the eleven innovative learning methods provided and for each give its benefits and challenges to their effective use.
- 5.2.4 For the argumentation and the computational learning methods, explain the stages involved and the elements that comprise it respectively.

5.2.1 Innovative Learning Methods

An innovative learning method is about solving given problems in a different way. This calls for creativity in the problem solving skills. Innovation therefore involves a different way of looking at a problem and solving it.

Modern Pedagogy Methods

Modern pedagogy is used to mean learner centred and activity based learning methods, which are used to get learners fully involved. This approach enables the student to construct his/her knowledge and skills through active participation in the teaching-learning process. The teacher acts as a guide, leading learners to achieve the objectives through the activities that the learners are engaged in during the classroom interactions. The following are some learning methods, in which the learner actively participates in the learning process. i Crossover learning ii learning

through argumentation iii Incidental learning iv Context based learning. v Computational learning vi Learning by doing vii Adaptive learning viii Collaborative learning ix Cooperative learning x Blended learning xi E-learning.

We shall now consider each of these innovative learning methods.

Crossover Learning Method

Crossover learning method is premised on the understanding that learning does not end or stop in the classroom. It involves the teacher proposing a question or problem in the class to be solved in the museum visit or a field trip. The student could tackle the task by taking down notes, collecting pictures, asking other people for their own thoughts, students can learn. The student can then present what he/she has learnt back to the classroom to further illuminate the given problem. The crossover learning methods has several benefits including enriching textbook knowledge with personal experience.

Learning Through Argumentation

Argumentation involves elaboration, reasoning and reflection. Which contribute to deeper conceptual learning. Argumentation is seen as a way of refining information and establishing the truth on an issue or to arrive at a common understanding. The focus is to arrive at a common goal. students are able to think critically and independently about important issues and contested values by arguing to learn rather than to quarrel.

In the argumentation process, reasoning to justify claims based on evidence or critical evaluation and revision with others is developed. When speakers take different positions, they challenge narratives, tend to make claims and give reasons and evidence to support them resolve the disagreement.

Argumentation is a powerful vehicle for reaching shared understanding. It is very important in research, and in science

Stages Of Sound Argumentation

Four stages of argumentation are given as

- i Claim
- ii The data
- iii The warrant and
- iv A qualifier

Incidental Learning

Incidental learning refers to any learning that is unplanned or unintended. It:

- i can develop while the student is engaged in a task or activity as a by product.
- ii can result from reflection on material that was consciously learnt but not recognized as relevant or useful at the time of the study.
- iii can happen when we least expect e.g. from talking with a friend or playing a video game.
- iv always happens in the context of another activity or experience

We can therefore see incidental learning as an indirect or additional (unplanned) learning which takes place in the context of an activity or experience.

Context Based Learning

Refers to learning whereby real life examples are used in a teaching environment so as to enable learning through actual/practical experience with a subject rather than using mere theoretical information. It is a learner centred approach to learning utilizing scenarios to replicate the social political context of the students or potential working environment.

Context based learning is premised on the understanding that learning takes place when the teacher is able to present information in such away that students can construct meaning based on their own experiences. Learners are therefore enabled to give meaning to concepts, rules and laws by creating connections between the concepts and relating them to their own experiences and the real world. Examples are such learning experiences as internship and service learning.

Computational Learning

Computational learning (also known as thinking and problem solving) is solving problems like a computer scientists. The computational learning is the thought process involved in formulating a problem and expressing its solution in such a way that a computer, human or machine can effectively carry it out. It is therefore a problem solving skill whereby, one looks at a problem and solves it systematically and arrives at a solution that both humans and computers can understand. Computational learning is made up of four elements as follows;

- i Decomposition
- ii Pattern recognition iii Abstraction iv Algorithm thinking

5.2.2 Learning By Doing Science

Learning by doing is a process of learning whereby students make sense of their experiences. The teachers seek to engage learners in more hands-on creative modes of learning.

The learning method is based on John Dewey's (1938) believe that development of practical skills was very crucial to children's education. Human beings therefore learn through a hands-on approach.

Asking questions systematically and objectively gathering data and testing hypothesis to find answers is the essence of scientific inquiry. In education, remote labs enables students to understand scientific principles through hands on laboratory experience 'learning by doing' from wherever they are located.

Remote Labs

Remote laboratories (also known as online laboratory or remote workbench) is the use of telecommunication to remotely conduct real (not virtual) experiment at the: i Real physical location of the operating technology, ii While the student is utilizing technology from a separate geographical location.

Remote laboratories are designed to allow students to remotely conduct real experiments across the internet. The actual laboratory, materials and the operating equipment are in one geographical location/place while the student is controlling the experiment at a different geographical location sometimes very far apart.

Though remote laboratory applications is relatively new in classroom teaching, Conducting experiments remotely has been in practice for many years. It has the potential to reach many students who are in remote locations and without laboratory resources.

Advantages And Disadvantages Of Remote Labs.

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Remote labs has the following advantages:

- i It enhances sharing of knowledge and experiences.
- ii It reduces startup costs of laboratories
- iii It improves learning outcomes to support better learning.
- iv It increases accessibility to laboratories.
- v It decreases fixed and variable expenditure.

However, remote labs have the following disadvantages:

- i Lack of equipment set up experience.
- ii lack of hands on trouble shooting experience.

5.2.3 Adaptive Learning.

Adaptive learning is a teaching – learning method which uses computer and other interactive teaching devices (algorithms) to orchestrate the interaction with the learner and deliver customized resources and learning activities to address the unique needs of each learner. (Wikipedia). Computers adapt the presentation of educational materials according to the students' learning needs as indicated by their responses to questions, tasks and experiences.

The technology encompasses aspects derived from various fields of study including computer science, education, psychology and brain science.

Adaptive learning is driven by the realization that tailored learning cannot be achieved on a large scale using traditional non adaptive approaches.

Adaptive learning endeavors to transform the learner from passive receptor of information to collaborator in the educational process.

The Importance Of Adaptive Learning.

Adaptive learning has several benefits some of which are summarized as follows; i Helps in providing focused attention on an individual. ii Provides for one on one instruction. iii Creates time efficiency

- iv Brings about confidence based approach
- v

Creates individualized learning paths.

- vi Creates personalized learning for a heterogeneous group.
- vii

Provides focused remediation.

Collaborative Learning Methods

This is a learning method that groups students to work together towards a common goal. The learning method is based on the collaborative learning theory which emphasizes that: a. knowledge is a social construct.

- b. Educational experiences involve interaction and social exchange.

- c. Contextually relevant and engaging learning activities are student centred and lead to deeper learning.

In addition, experts claim that:

- i the active exchange of ideas within groups of students promotes critical thinking.

- ii Teams that engage in cooperative learning achieve; o Higher levels of thought and o Retain information longer than students who work solely as individuals.

Strategies For Effective Collaborative Learning

Collaborative learning is hard work which does not just happen. It has to be intentionally designed so as to be effective. There are strategies to promote effective collaborative learning.

8.12.2 Advantages Of Collaborative Learning.

This method helps students learn how to work effectively as a team. Cooperative learning is a specific kind of collaborative learning where students work together in small groups on a structured activity.

The students are individually accountable for their work and the work of the group as a whole is also assessed. Cooperative groups work face to face and learn to work as a team.

5.2.4 Cooperative Learning

Cooperative learning is where students of mixed levels of abilities are arranged into groups and rewarded according to the group's success rather than the success of an individual member. Sometimes cooperative learning method is thought of simply as group work but groups working together might not be working collaboratively. Cooperative learning is characterized by five elements as follows:

- i Face to face interaction
- ii positive interdependence
- iii individual accountability
- iv group processing
- v collaborative skills

Types Of Cooperative Learning

There are three types of cooperative learning. The teacher may use one or more types of groups at a time. The three types are as follows:

- i Informal learning groups
- ii Formal learning groups
- iii Cooperative base groups

The Benefits And Limitations Of Cooperative Learning

There is no one best learning method for all learning environments. The Teacher has to be aware of all the available learning methods then select the most suitable or combine several learning methods depending on the concepts being taught, the students learning needs and the instructional media available. The cooperative learning methods has many benefits to the learner but also has some limitations.

Blended Learning

This refers to learning which combines technology (online) and traditional learning modes. It therefore combines the strengths of face to face and online learning by extending the reach of the instruction beyond the classroom through the use of digital resources. For example, the instructor may ask students to view a provided video before he/she introduces the concept in class.

Benefits And Challenges Of Blended Learning

Blended learning combines the strengths of both face to face and the digital learning. The benefits can be summarized as follows: i Its both efficient and effective in content presentation

- ii Its efficient and quick in reaching a big audience
- iii it saves on travel to attend classes and missing work and therefore it is cost effective.
- iv It provides desirable flexibility in presenting the content.
- v It seamlessly transitions from classroom to computer and vice versa when well used.
- vi It covers all learning styles through use of well chosen mediums and techniques.
- vii It compensates for limited classroom space.

On the other side, there are several barriers or challenges to blended learning which can be summarized as follows:

- i expense of technology
- ii Inadequate training
- iii Technological issues
- iv The need to adapt content to blended learning
- v Weakened relationship between the teachers and the students.

E-Learning

There are many terms used to describe e learning such as distance learning, computerized electronic learning, online learning and internet learning. We can describe e-learning as learning that utilizes electronic technologies to access educational content outside the traditional classroom. It can also be described as the acquisition of knowledge which takes place through electronic technologies and media.

e-learning is interactive (the student can raise his/her hand and interact with the teacher or other students in the class) and therefore use of video tape, TV channel, do not facilitate learning. E learning is conducted on the internet where students can access their learning materials online at any place and any time.

Challenges To Use Of E-Learning

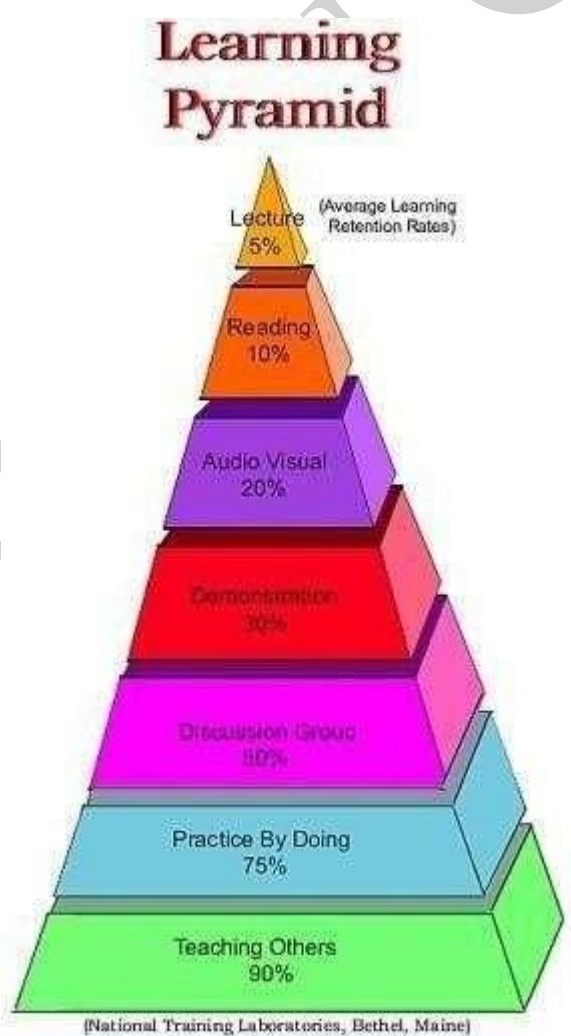
The interactive nature of the learning process such as viewing a video, listening to audio and reading through a manual makes the content very interactive and easier to recall information.

However, e- learning has its share of challenges including the following:

- i limited funds
- ii poor internet access in some areas
- iii lack of enough trained staff

Numbering, pacing & sequencing	5.2.2
Title	Innovative learning methods
Purpose	The purpose of this e-tivity is to enable you to understand a wide range of the 21 st century teaching-learning methods that facilitate competence based learning and enable you select the most suitable to use depending on learners' needs, content complexity and the objectives to be achieved.

Brief summary of the overall task	□ Using the content in the provided link and https://asaolusan.wordpress.com and https://issuu.com, and https://asaolusan.wordpress.com i Explain why innovative learning methods have to replace the traditional learning approaches. ii Define innovative learning in your own words. iii Explain why modern pedagogy methods are necessary in the 21st Century iv For each of the discussed innovative learning method, explain the benefits & steps to be followed for their effective use.
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Spark	 Learning Pyramid (Average Learning Retention Rates) <table border="1"> <thead> <tr> <th>Learning Method</th> <th>Retention Rate</th> </tr> </thead> <tbody> <tr> <td>Lecture</td> <td>5%</td> </tr> <tr> <td>Reading</td> <td>10%</td> </tr> <tr> <td>Audio Visual</td> <td>20%</td> </tr> <tr> <td>Demonstration</td> <td>30%</td> </tr> <tr> <td>Discussion Group</td> <td>50%</td> </tr> <tr> <td>Practice By Doing</td> <td>75%</td> </tr> <tr> <td>Teaching Others</td> <td>90%</td> </tr> </tbody> </table> (National Training Laboratories, Bethel, Maine)	Learning Method	Retention Rate	Lecture	5%	Reading	10%	Audio Visual	20%	Demonstration	30%	Discussion Group	50%	Practice By Doing	75%	Teaching Others	90%
Learning Method	Retention Rate																
Lecture	5%																
Reading	10%																
Audio Visual	20%																
Demonstration	30%																
Discussion Group	50%																
Practice By Doing	75%																
Teaching Others	90%																

Individual task	□ Using bullet points i Compare the teacher's and the student's roles in traditional learning approaches and the modern innovative learning approaches. ii In one of your teaching subjects, identify a learning situation in a specified topic, when it would be advisable to use one of the innovative learning methods presented in this lesson (repeat this task for all the other innovative learning methods) iii Using the provided links https://www.teachingcommons.stanford.edu and https://en.wikipedia.org and identify a topic in any of your teaching subjects which can be best learnt through experimentation and which can be done through a remote lab. (Do i, ii & iii on the discussion forum 1.3 on the kusoma platform)
Interaction begins	Provide your assessment on the suitability of the topic given by the team learners in which experimentation would be done through remote lab method. Do this on the discussion forum 1.3 on the Kusoma platform
E-moderator interventions	1. Ensure that learners are focused on the contents and context of discussion. 2. Stimulate further learning and generation of new ideas. 3. Provide feedback on the learning progress. 4. Close the e-tivity.
Schedule and time	This task should take three hours
Next	

References

1. A. Mousa and A. Mubarak, *E-learning, Foundations and Applications*, Data Network Corporation, Riyadh, Saudi Arabia, 2005.
2. A. M. Al-Karam and N. M. Al-Ali, "E-learning: the new breed of education," in *Education development through utilization of technology: UNESCO Regional Office for Education in the Arab States*, V. Billeh and A. Ezzat, Eds., pp. 49–63, 2001. View at: [Google Scholar](#)

3. H. Salama, *Blended learning: Natural Evolution for E-learning*, *E-learning Forum*, 2005, <http://www.elearning.edu.sa/forum/attachment.php.attachmentid=635&d=118206521>.
4. D. R. Garrison and H. Kanuka, "Blended learning: uncovering its transformative potential in higher," *The Internet and Higher Education*, vol. 7, no. 2, pp. 95–105, 2004. View at: [Publisher Site](#) | [Google Scholar](#)
5. M. Khalafullah, "Effectiveness of the use of e-learning and blended education in the development of the skills of the production of educational models in the students of the Department of Educational Technology, Faculty of Education, Al-Azhar University," *Journal of the Faculty of Education*, 2010. View at: [Google Scholar](#)
6. J. Watson, *Blended Learning: The Convergence of Online and Face-To-Face Education: Promising Practices in Online Learning*, North American Council for Online Learning, Vienna, VA, USA, 2008.

Questions

1. Explain the teacher's role in a learning through argumentation situation.
2. Justify the context-based learning as an innovative learning method.
3. Discuss the elements of computational learning
4. Why is learning by doing an important learning method?
5. Based on a topic in one of your teaching subjects, give a problem which requires computational learning skills to solve it.

E Resources

1. www.fao.org
2. <https://e-student.org>
3. www.elearningnc.gov
4. <https://www.knowledgewave.com>
5. <https://study.com>,
6. <https://www.education.vic.gov.au.com>
7. <https://www.education.vic.gov.au.com>
8. <https://info.getadministrative.com>
9. <https://info.getadministrative.com>
10. <https://www.edutopia.org/video>
11. <https://www.thirteen.org>
12. <https://www.edutopia.org>
13. <https://en.m.wikipedia.org>

LESSON 6: CHANGING TRENDS IN TEACHER EDUCATION

Professor John N. Maundu

6.1 Introduction

Human beings are dynamic in various spheres of their existence. In education, humankind is constantly looking for ways to improve teaching and learning engagements. Through research and social interaction, various strategies and techniques are often generated to improve teacher education in general and classroom instruction in particular. Various policies and instructional practices are established to ensure that teachers, learners and relevant stakeholders at large are updated on what is in vogue.

In keeping abreast with what is in tandem with modern day goals of education, the educator in general and the classroom lecturer in particular, needs to understand both past practices and future aspirations. This way, appropriate focus will be maintained as the school, college or university seeks to uphold expected performance standards of the learners.

6.2 Learning Outcomes

By the end of this lesson you will be able to:

- 6.2.1 Identify key trends in teacher education
- 6.2.2 Create interactive learning environments for prospective teachers
- 6.2.3. Develop and use face-to-face and hybrid digital pedagogies
- 6.2.4 Use Open Education Resources in teaching and learning environments
- 6.2.5 Plan and implement Large class pedagogic strategies in instruction

6.2.1 Key trends in Teacher Education

Teacher education has for many decades been characterized by centralization of educational programmes and authoritarian delivery of content. In countries throughout the world, the management of education has been under a specific body or ministry that provides the curricula to be taught with minimum learner input. With time however, demands for more inclusive curriculum development and implementation strategies have emerged, paving way to democratic participation in various matters of teacher training and classroom teaching. The current practice of integrating technology in classroom instruction is one such contemporary trend among several others. This should embrace the 21st Century skills, such as creativity, critical thinking and collaboration.

E-tivity 6.2.1 Key trends in teacher education

Numbering, Pacing and Sequencing	6.2.1
Title	Key trends in teacher education
Purpose	The purpose of this e-tivity is to enable you state examples of current and emerging trends in teacher education
Brief summary of overall task	Open these links to recognise key trends in teacher education <u>https://kitaboo.com/trends-in-education-technology/</u> <u>https://www.gettingsmart.com/2017/07/5-emerging-trends-in-21st-century-education/</u> <u>https://wabisabilearning.com/blogs/literacy-numeracy/skills-every-student-needs</u>

ODHIAMBO DARREN GREGORY

+254704597085

Spark	<i>New wine in old skins....No way!</i>
Individual task	1. State at least three key trends in teacher education 2. Find another 21st Century skill link and state two skills that are not in the link above
Interaction begins	post your list of the trends on the <i>kusoma discussion forum</i> ; indicate is your response to the feedback that will be given thereafter 6.2.1
E-Moderator interventions	1. The statements should be related to the to the Kenyan context 2. Discussion on the statements undertaken 3. Close the e-tivity
Schedule and time	Two hours should be adequate for the above tasks
Next	Creating interactive learning environments for prospective teachers

6.2.1 References

1. Jim Ackers (2018). Promoting inclusive teacher education: methodology.
2. UNESCO, *Teacher Education & Practice*, Vol. 30, No. 1, ISSN 0890-6459

6.2.2 Creating interactive learning environments for prospective teachers

Traditional teaching and learning (Web 1.0) has largely been characterised by a lot of teacher talk and passive learning. Modern day instruction (Web 2.0) emphasizes learner-centred pedagogy where collaborative, multisensory and proactive learner participation is the norm. Here the use of interactive strategies that encourage constructivism on the part of the student is a key feature of the instructional process.

E-tivity 6.2.2 Creating Interactive learning environments

Numbering, Pacing and Sequencing	6.2.2
Title	Creating Interactive learning environments
Purpose	The purpose of this e-tivity is to familiarize you with the concept of interactive learning environments
Brief summary of overall task	The video https://study.com/academy/lesson/types-of-learning-environment.html enables you to list and explain four types of learning environments, the link

Spark	 <i>Can you shape the future of education?</i>
Individual task	From the pdf link,\ above, explain 3 roles of the teacher and 3 roles of the student in actualizing interactive learning
Interaction begins	Draw a flow diagram linking the Pedagogy, constructivism and search for Web 2.0 tools in an interactive classroom. Post the diagram on the <i>kusoma discussion forum</i>
E-Moderator interventions	1. Web 2.0 tool or tools in the diagram to be identified 2. Feedback on diagram to be given 3. Close the e-tivity
Schedule and time	Approximately 2 hours should be adequate for the above tasks.
Next	Face-to-face and hybrid digital pedagogies

6.2.2 Reference

1. Roxana Moreno & Richard Mayer (2007).Interactive Multimodal Learning Environments Special Issue on Interactive Learning Environments: Contemporary Issues and Trends. *Educ Psychol Rev* (2007) 19:309–326
2. Hsiu-Mei Huang a , Ulrich Rauch b , Shu-Sheng Liaw c ,*(2010).Investigating learners' attitudes toward virtual reality learning environments: Based on a constructivist approach *journal homepage: www.elsevier.com/locate/compedu*

6.2.3 Development of Face-Face and Hybrid digital pedagogies

In Africa and many other parts of the world, Face –to-Face teaching has been the normal pedagogical mode of content delivery. However, from the 20th Century, advances in technology have influenced the way teachers implement the curriculum in the classroom. One significant step in using and integrating technology in education is *Hybrid* teaching.

The *hybridity* mechanism involves the use of **two** dissimilar teaching pedagogies to yield a result that has advantages (and limitations) from both methodologies. In this effort, the delivery of the content is split into *online* and *face-to-face* portions. The teacher is supposed to borrow the best ideas from each instructional arena. Hybrid learning is also referred to as *Blended* learning.

E-TIVITY 6.2.3 FACE-TO-FACE AND HYBRID PEDAGOGIES

Numbering, Pacing and Sequencing	6.2.3
Title	Face-to-Face and Hybrid pedagogies

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+254704597085

Purpose	By the end of this e-tivity you will be able to use a hybrid teaching method
Brief summary of the overall task	The links below will enable you to distinguish Face -to Face from Hybrid teaching. Upon going through the two links, you will be challenged enough to create and execute a hybrid pedagogy in your lessons. https://untblendedlearning.weebly.com/what-is-blended-learning.html https://files.eric.ed.gov/fulltext/EJ1120647.pdf https://cft.vanderbilt.edu/guides-sub-pages/blended-and-online-learning/
Spark	TRUE or FALSE?: Face- to- Face +Hybrid = 2 lesson plans
Individual task	1. In your own words, define hybrid method of teaching 2. State two changes that need to be made in order to embrace hybrid learning 3. State a challenge in using hybrid instruction and suggest a possible solution
Interaction begins	Post answers to the task above in the <i>kusoma discussion forum</i>
E-Moderator interventions	1. Identification of Web 2.0 tools that could be used to overcome specific face –to-face limitations 2. Provision of feedback on the discussion forum 3. Close the e-tivity
Schedule and time	Accomplish the above tasks in 2 hours
Next	Use of Open Education Resources in teaching and learning environments

6.2.3 References

1. Karl Wuensch, Erol Ozan, MasaoKishore (2006). Pedagogical Characteristics of Online and Face- to-Face Class
2. Bonk, C. & Graham, C. (2005). *Handbook of blended learning: Global perspectives, local designs*. San Francisco, CA: Pfeiffer Publishing.
3. Bonk, C., Kim, K. J., & Zeng, T. (2006). Future directions of blended learning in higher education and workplace learning settings. In C. Bonk & C. Graham (Eds.), *the handbook of blended learning: Global perspectives local designs* (pp. 550-567). San Francisco: Pfeiffer.
4. Garnham, C., & Kaleta, R. (2002, March). Introduction to hybrid courses. *Teaching with Technology Today*, 8 (10).

6.2.4 Using Open Education Resources in teaching and learning environments

Open Education Resources (OER) are a collection of digital technology materials that are publicly available online for educational purposes. They are published under open licenses i.e. anyone can access them and use them as they are or modify them to suit specific needs in education. In teaching and learning, OERs could be in the form of an entire course programme, lecture notes, simulation videos, and student assignments among others. OERs are also known as *Creative Commons*. It is of paramount importance for prospective and practicing lecturers to know how to use these resources.

E-tivity 6.2.4 Using OERs in teaching and learning environments

Numbering, Pacing and Sequencing	6.2.4	
Title	Using OERs in teaching and learning environments	
Purpose	This e-tivity seeks to enable you to state and explain ways in which you can use OER materials in instruction	
Brief summary of the overall task	This video link https://www.youtube.com/watch?v=CJHDr3s0V7M Provides you with some ideas on how to use open access materials in your classroom.	
Spark	 No restrictions to learning	
Individual task	1. After watching the above video, identify three ways in which you could source and use OER in one of your teaching subjects 2. Google to find a list of OERs	
Interaction begins	Post on the <i>kusoma discussion forum</i> one way of sourcing and one way of using a Creative Common material in instruction	
E-Moderator interventions	1. Relate the above post to the Kenyan context 2. Close the e-tivity	
Schedule and time	About 2 hours will be sufficient to complete the above tasks.	
Next	Planning and implementing large class pedagogic strategies	

6.2.4 References

1. <https://pitt.libguides.com/openeducation/biglisthttps://raccoongang.com/blog/quick-list-open-educational-resources-examples/>
2. <https://www.opencolleges.edu.au/informed/features/10-open-educational-resources-know/>

6.2.5 Planning and implementing large class pedagogic strategies in inclusive classrooms The classroom teacher as the implementer of the curriculum is beset with many challenges. . One such challenge is on how to undertake meaningful delivery of content to a large number of students. While it is true that there is no universal definition of the concept of ‘large classes’, it is generally agreeable that any class size that a teacher cannot manage effectively and engage the learners actively, may be considered as *large*. In this regard, inadequacy of teaching and learning resources

is a crucial contributor to the level of learning effectiveness as students will not be adequately engaged. Further, a teacher's work experience and socio-cultural background will play a major role in how he/she manages large classes.

Despite the differences among teachers on how they create and manage learning environments, it is necessary that all teachers become aware of some of the pedagogic strategies that have been found to work well in various parts of the world including Kenya, for improved learner benefits from the education system.

E-tivity 6.2.5 Planning and implementing large class pedagogic strategies

Numbering, Pacing and Sequencing	6.2.5
Title	Planning and implementing large class pedagogic strategies
Purpose	This e-tivity will facilitate you to identify various pedagogic strategies available to a classroom teacher/lecturer
Brief summary of overall task	Open these two links to learn about teaching large classes: http://breda-guide.tripod.com/New-5.htm https://cft.vanderbilt.edu/guides-sub-pages/teaching-large-classes/
Spark	
Individual task	List and explain 3 strategies that are applicable to a large class teaching and learning environment at university level
Interaction begins	Discuss your strategies with another student taking this course and post common viewpoints and dissimilarities on the <i>kusoma discussion forum</i>
E-Moderator interventions	1. The 3 strategies listed above should be applicable within the Kenyan context 2. Closing the e-tivity
Schedule and time	2 hours should be adequate for the tasks above
Next	Further individual reading from a variety of textbook and internet sources.

6.2.5 References

1. Gathumbi A., Bwire A. (Eds.) (2018). Building capacity through quality teacher preparation: A multifaceted approach. Printed by Kenyatta University Press.

ODHIAMBO DARREN GREGORY

+254704597085

2. <https://www.teflcourse.net/blog/advantages-and-disadvantages-of-teaching-large-classes/>
3. <https://www.insidehighered.com/advice/2017/09/19/how-teach-large-classes-students-effectively-essay>

6.2.5 Assessment Questions

1. Which one of the following is in 2020, a common practice in teacher education in Kenya?
 - A. Use of mobile phones by students in learning at school
 - B. Integration of technology in classrooms by teachers
 - C. Use of Google Lens in classroom instruction
2. Collaborative learning means _____
3. Circle the letter bearing the correct answer in the question below
The major purpose of Hybrid pedagogy is A) involve the teacher in content delivery B) introduce constructivism in learning C) take advantage of what is good in online and face- to-face
4. Which ONE of the following is NOT true about use of OERs in instruction?
 - A. They facilitate modification of content
 - B. Teachers require operating licenses
 - C. They facilitate learning at own pace
 - D. A forum for seamless sharing of content is available
5. State **three** advantages and **three** limitations of teaching a large number of students

LESSON 7

MOTIVATION AND SUSTAINING ATTENTION IN THE CLASSROOM (Dr. Mary W. Nasibi)

7.1 Introduction

In this lesson, we will focus, on the concept of motivation, the types of motivation, the theories of motivation and their application to education. We will also learn about how to motivate and sustain learners attention in the classroom.

7.2 Lesson learning outcomes

By the end of this lesson, you will be able to:

- 7.2.1 Explain the concept of motivation.
- 7.2.2 Explain the types of motivation
- 7.2.3 Explain the importance of motivation in teaching and learning.

7.2.4 State the theories of motivation

7.2.5 Explain Maslow theory of motivation and its relevance to education

7.2.6 Explain ways in which you can motivate and sustain motivation in the classroom.

7.2.1 The concept of motivation

Motivation is essential for effective teaching and learning. As a concept it can be defined as the reason why people behave the way they do or do whatever they do. According to Woodfolks 91992) in Nasibi(2003) motivation is the general process by which behaviour is initiated and directed towards a goal.

There are two types of motivation which explain learner's behaviour in the classroom: intrinsic and extrinsic motivation. Intrinsic motivation is influenced by the reinforcers that are present in the activity being performed such as joy and satisfaction to the performer. Extrinsic motivation is influenced by outside rewards to the action being performed.

E-tivity 8.2.1 – Concept of motivation

7.2.2 Importance of motivation in teaching and learning

Educators have argued that motivation is part and parcel of learning because

; 1. It stimulates and makes learning interesting and sustainable .

Numbering, pacing and sequencing	8.2.1
Title	Definitions of motivation , intrinsic and extrinsic motivation
Purpose	The purpose of this e-tivity is to enable you explain the meaning of motivation and to be able to between intrinsic and extrinsic motivation.
Brief summary of overall task	Watch videos1 and2 on these links; http://www.edpsycinteractive.org/topics/motivation/motivate.html http://www.edugyan.in/2017/03/motivation-definition-classification.html (Website)

2. It increases concentration and alertness among the learners.

Spark	
Individual task	a) Using bullet points, outline the key themes in the definitions of motivation b) In a sentence develop your own definition of motivation c) In one statement differentiate intrinsic and extrinsic motivation.
Interaction	a) Post two themes that are common in the definition of motivation b) Provide positive and constructive feedback on the terms defined .
E-moderator interventions	1, Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas 3. Provide feedback on the learning progress 4. Close the e-tivity
Schedule and time	This task should take one hour.
Next	

3. It improves learners performance both in class tasks and in examination.

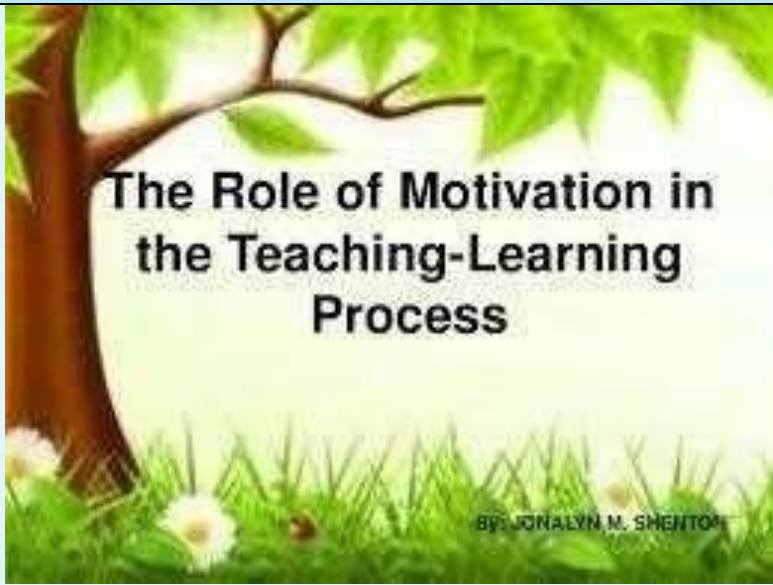
4. It improves creativity and initiative among the learners.

E-tivity 7.2.2 The importance of motivation

Numbering, pacing and sequencing	8.2.1
Title	Importance of motivation in teaching and learning
Purpose	The purpose of this e-tivity is to enable you to discuss the importance of motivation

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Brief summary of overall task	Watch videos1 and2 on these links; https://study.com/academy/lesson/the-importance-of-motivation-in-an-educational-environment.html https://theeducationhub.org.nz/motivation/ (Website)
Spark	
Individual task	a) Using bullet points, outline the key themes in the importance of motivation b) In two sentences explain what is likely to happen in a class where the teacher is not aware of the benefits of motivation. .
Interaction	a) Post two themes that are common in the importance of motivation b) Provide positive and constructive feedback on the points given. .
E-moderator interventions	1,Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas 3. Provide feedback on the learning progress 4. Close the e-tivity
Schedule and time	This task should take one hour.
Next	The theories of motivation

7.2.3 Theories of motivation

Motivation theories though many and varied can be categorized in to behaviourist, humanist, cognitive and social learning theories. The advocate of behavioural was B.F Skinner (1938) argued that all behaviours are motivated by rewards. In a learning situation therefore a learner has to be rewarded for making a positive response. This will propel him/ her to move to the next level.

Among the proponents of humanistic theory are Ibrahim Maslow(1943)\and his well known theory of hierarchy of needs. Perumal (2009) posits that in this theory “ the single, holistic principle that binds together the multiplicity of human motives is the tendency for a new and higher need to emerge as the lower need fulfils itself by being sufficiently gratified.”

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Numbering, pacing and sequencing	8.2.3
Title	The theories of motivation
Purpose	The purpose of this e-tivity is to enable you to discuss theories of motivation
Brief summary of overall task	Watch videos1 and2 on these links; https://www.simplypsychology.org/maslow.html https://www.researchgate.net/publication/30 (Website) https://courses.lumenlearning.com/boundless-psychology/chapter/theories-of-motivation/ (Boundless)
Spark	
Individual task	a) Using bullet points, outline the key features in Maslow theory of motivation and any other t b) In a sentence for each of the theories in(a) explain how the theory can be applied in educati

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Herzberg's motivational theory argues that if there is improved learning environment, communication, administration procedures and resources then there will be enhanced positive motivation among learners. Expectancy theory is differentiated under cognitive theory and motivation is enhanced when there is a likely hood of success in the activity or task at hand, Related to this is expectancy-valued theory which states that motivation is influenced by the reason for being involved in the task. **E-tivity**

7.2.3 Theories of motivation

7.2.4 Motivation techniques in the classroom

Motivation in the classroom is influenced by the learning environment, the teacher's experience and expertise in motivation techniques and the learners' nature of motivation whether they are intrinsically or extrinsically motivated.

A conducive learning environment which is free of destructors, neat, pleasant, comfortable, bright, with

Interaction	a) Post two themes that are common in Maslow theory of motivation b) Provide positive and constructive feedback on the points given. .
E-moderator interventions	1.Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas 3. Provide feedback on the learning progress 4. Close the e-tivity
Schedule and time	This task should take one hour.
Next	The concepts teaching and learning

attractive and relevant displays is a plus for learners motivation (Nasibi 2003).

The techniques to be put in place include reinforcement techniques, stimulus variation, set induction and introducing a variety, novelty and familiarity in the lessons.

It is important that the teacher maintains success expectation, gives learners opportunity for active response, uses interesting methods, caters for individual differences among the learners. Besides, the educator has to use students present motives, minimize performance anxiety and give immediate feedback.

E-tivity 7.2.4 Motivational techniques

Numbering, pacing and sequencing	8.2.4
Title	Motivation techniques in the classroom
Purpose	The purpose of this e-tivity is to enable you to discuss ,how you can motivate learners in a learning environment
Brief summary of overall task	Watch videos1 and2 on these links; https://www.nbpts.org/top-5-strategies-for-motivating-students/ (Top 5 Strategies for Motivating Students...) https://ucat.osu.edu/bookshelf/teaching-topics/learning-about-learning/techniques-motivate-students/ (Techniques to Motivate Students - UCAT)
	https://www.teachthought.com/pedagogy/21-simple-ideas-to-improve-student-motivatio/ (Techniques to Motivate Students - UCAT ; 21 Simple Ideas To Improve Student Mo...)

Spark	
Individual task	a) Using bullet points, outline the key themes in motivation techniques b) In a sentence for each of the techniques give examples .
Interaction	a) Post two themes that are common in Maslow theory of motivation b) Provide positive and constructive feedback on the points given. .
E-moderator	1, Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas 3. Provide feedback on the learning progress 4. Close the e-tivity
Schedule and time	This task should take one hour.
Next	Sustaining learners attention

7.2.5 Sustaining motivation

Once motivation has been achieved, the instructor has to ensure it is sustained throughout the lesson. Psychologists argue that the students' attention span is short lasting between 10 to 15 minutes on the average. For learning to go on, the teacher has to come up with strategies of maintaining the motivation.

Below are some of the techniques one can put in place to ensure learning goes on:

1. Use of set induction at transition or evaluation stages of the lesson. This can be achieved by use of paradoxical statements, presentation of novel stimuli, use of behavioural surprise etc

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2. Applying stimulus variation through focusing (use of verbal statements, specific gestures, introduction of props, emphatic movements)
3. Utilisation of reinforcement techniques where desirable behaviour is reinforced and undesirable is discouraged through punishment.
4. Use of examples which are simple, relevant, interesting and dramatic.,
5. Establish incentive contingencies.
6. Provide immediate feedback.

E-tivity 7.2.1 Sustaining attention

Numbering, pacing and sequencing	8.2.1
Title	Sustaining attention in the cclassroom
Purpose	The purpose of this e-tivity is to enable you to discuss ,how you can sustain learners attention in class
Brief summary of overall task	Watch videos1 and2 on these links; https://facdev.e-education.psu.edu/ol2000/lesson2p7 (Techniques to Motivate Students - UCAT ; 21 Simple Ideas To Improve Student Mo...; Strategies for Sustaining Students' M...) https://learningsolutionsmag.com/articles/1552/seven-tips-to-sustain-motivation-and-keep-learners-coming-back-for-more (Techniques to Motivate Students - UCAT ; 21 Simple Ideas To Improve Student Mo...; Strategies for Sustaining Students' M...; cory-mcmillen 2014)
Spark	

Individual task	a) Using bullet points, outline the key themes covered by motivation techniques b) In a sentence explain each of the techniques stated in the notes. Give examples .
Interaction	a) Post two themes that are common in Maslow theory of motivation b) Provide positive and constructive feedback on the points given. .
E- moderator	1, Ensure that learners are focused on the content and context of the discussion 2. Stimulate further learning and generation of new ideas 3. Provide feedback on the learning progress 4. Close the e-tivity
Schedule and time	This task should take one hour.
Next	Providing for individual differences.

Assessment Questions

1. With examples differentiate between intrinsic and extrinsic motivation.
2. Discuss how Maslow theory of hierarchy of needs is relevant to education.
3. What is the relevance of social theory to classroom teaching and learning.
4. Explain how a teacher can motivate an intrinsically student in his/her class.
5. Discuss with examples three strategies a teacher can put in place to sustain the learners attention in class.

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LESSON 8

EFFECTIVE CLASSROOM COMMUNICATION

PROF.J.N.KIMEMIA

8:1 Introduction

Communication is a unique human activity that we are all involved in. They say communication is the life- blood of any group of people or organization including the family members, church, schools ,communities and nations. Human beings live socially because of their ability to communicate.Lack of communication among people is like the HIV virus or Corona (Covid-19) disease.Language is the principal tool of communication but we are to communicate without using language. We may communicate using gestures, symbols, visual forms, written messages, electronic devices, body movements or even manner of dressing as well as remaining silent or mute. Teachers and all other professionals must communicate effectively in order to achieve their teaching and learning goals. This is what is probably known as interpersonal communication .It takes place between the teachers and learners as well as among the learners themselves.

The communication process is vital in the classroom teaching and learning process in the classroom. Effective communication will lead to effective teaching which will in turn lead to effective learning.

8.2 Learning Activities and Outcomes

At the end of the lesson or lessons the learners will be able to:

- 8.2.1 Explain the communications process using the SMCR model.
- 8.2.2 Explain the main qualities of a teacher in classroom control and management.
- 8.2.3 Identify main barriers to communication
- 8.2.4 Apply questioning and feedback methods in classroom communications.

8.2.1 Definition of Communication

There are many and varied definitions of communication .Historians, sociologists ,engineers, technologists and other professionals use the term communications differently depending on their contexts and aspects referred. In our teaching and learning context we shall confine the term communication to passing information or ideas from the teacher to the learners as well as receiving the feedback from the learners in the classroom. There are three aspects to be considered in the communication process. These include:

- i. That, communication involves **people** .This implies that communication tries to understand how people relate to each other.

- ii. That, communication involves **shared meaning** implying that for effective communication there must be agreement in words and terms used in the communication .This is also called **mutuality** of meanings.
- iii. That communication may be symbols gestures sound letters numbers and words that represent or approximate ideas that are meant to be communicated.

Forms or Types of Communication

In transmitting information or messages from one person to another the following forms or types of communication could be used:

- i. Written form
- ii. Oral (verbal) form.
- iii. Pictorial (visual) form.
- iv. Gestures, body movements including hands, face, mouth, eye, feet or body posture.
- v. Manner of dress.
- vi. Silence or mute.

Process of Communication (Concepts in Communication)

The available literature highlights the process of communication consisting or comprising

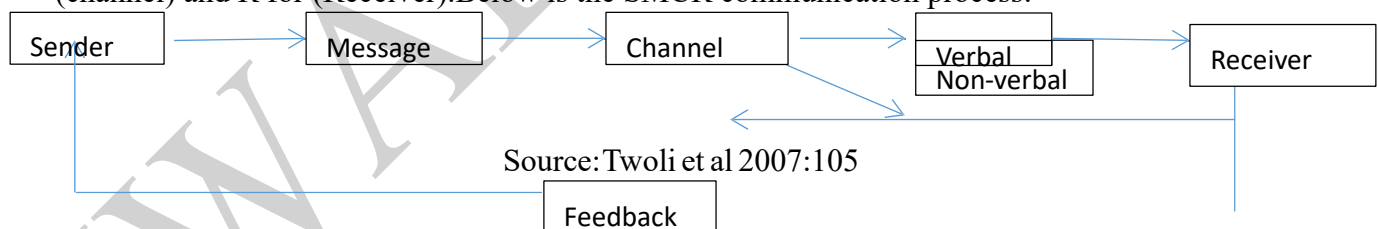
- i. Sender
- ii. Message
- iii. Receiver.

The figure below illustrates the three components or parts of the communication process.

A MODEL OF COMMUNICATION PROCESS

Sender → Message → Receiver

For effective communication the Sender initiates the message to be transmitted .The Sender **encodes** the information with the conviction that there is mutuality or shared meanings between the Sender and the Receiver. In turn, the receiver decodes the information or message and responds to the sender. Between the Sender (source) and the receiver there can be other modes of transmitting the messages. This is called the **channel** which may send the message in various forms e.g. Verbal and Non-verbal .The receiver should give the **feedback** to the sender for effective communication to take place. Schematically this is referred to as SMCR model where S stands for (source) M for (message) C for (channel) and R for (Receiver).Below is the SMCR communication process:



In the classroom the teacher initiates the communication process the teacher is the source. The learner is the receiver of the message or information. The teacher asks questions to the learners. The students answer the questions. The channel may be verbal or Non-verbal. The answers the questions are the feedback in the communication process.

8.2.2 Concepts in Verbal and Non-verbal Communication

Verbal communication is the most commonly used channel by teachers in the classroom. For effective verbal communication the following aspects should be considered:

- i. The teacher should use clear and audible voice to be heard in the whole classroom .The teacher should not shout or speak in too low or soft voice. The voice should be appropriate so that student's listen comfortably to be able to be assimilate the information given by the teacher.
 - ii. The pitch of the teachers' voice should be appropriate or moderate .The teacher should vary the pitch or tone according to the needs of the classroom and the learners. iii. The voice clarity is also important in verbal communication pronunciation problems may negatively affect the communication of the verbal communication of the teacher.
 - iv. The teacher should face the learners for effective verbal communication to take place.
- B. Non-verbal communication is also common in classroom interaction with the learners. These include the use of gestures, facial expressions, eye contact, use of hands and body movements' .These may send special messages from the teacher to the learners. Both the teacher and the learners may successfully utilize non-verbal communication in the classroom teaching and learning process.
- C. To improve and compliment classroom communication the teacher should plan and utilize the writing board blackboard, whiteboard and / smart board. These devices should be used to reinforce and clarify oral communication .The following steps are important in improving teaching and learning:
- i. Plan the teaching and use of boards in advance
 - ii. Ensure that what is written on the boards is neat, tidy and legible by all learners in the classroom. iii. Write clear and large words phrases and sentences on the board. iv. Clear the boards after use do not write all over the board and avoid confusing the learners.

E- Activity

- i. **Title** effective communication in classroom
- ii. **Purpose**; stimulate and focus on importance of communication for teachers and learners
- iii. **Brief summary** – Highlight importance of communication of overall task in teaching and learning in classroom verbal and non-verbal communication.
- iv. **Spark** - introduce communication with the metaphor “ He who communicates is he who leads”
- v. **Individual contribution**: ask questions on components or parts of communication process.
Source – Message- Receiver vi. **Interaction begins** – simple oral questions on communication
- vii. **E-moderator interventions** – factors that lead to effective communication – trust, carity, motivation, interest and benefits.
- viii. **Schedule and time** – one or two lessons will be adequate
- ix. **Next-** Barriers or hindrance to effective communication

8.2.3 Barriers to communication

There are various conditions that restrict or hinder effective communication in the classroom. These are:

- i. **Physiological barriers.** These are factors that affect our bodies, both for teachers and learners. They include sickness, deafness, blindness or any impairment.
- ii. **Psychological barriers:** These include attitudes, interests and expectations of either teachers or learners.
- iii. **Physical (environmental) barriers:** Noise and other natural conditions may hinder effective communication. Even the physical distance between the sender and the receiver may negatively affect effective communicating in the classroom.

Others barriers to effective communication include the following;

- i. Language differences
- ii. Lack of feedback
- iii. Lack of trust
- iv. Social distance or status
- v. Complexity of information
- vi. Quantity of information
- vii. Fear and anxiety
- viii. Lack of common experience
- ix. Gestures
- x. Prosodic features e.g. stress and intonation
- xi. Differing perceptions
- xii. Emotionality e.g. anger, love, defensiveness, hate, fear, jealousy, embarrassment
- xiii. Noise any factor that disturbs, confuses or interferes with the communication.
- xiv. Technical jargon – Too many assumptions too long chains of command, wide span of control, inaccurate judgment or failure to communicate at all.

8.2.4 Effective Communication

This will be promoted and enhanced by:

- i. **Trust-** in classroom trust by teachers and learners will lead to effective communication and effective management of the classroom.
- ii. **Active listening:** Good and active listening demands “paying attention” to teachers and learners before responding on giving feedback.
- iii. **Feedback** – Without feedback communication it is incomplete and not effective
- iv. **Effective communication-** demands to have the right messages getting to the right people at the right time.
- v. There are desirable **channels** of communications either upwards, downwards and sideways. These channels should be used for effective communication ever in the classroom control and management.
- vi. Effective communication will lead to effective management and control of the classroom which will in turn lead to effective accomplishments of the goals and objectives of the lesson. This the **means** and **end** to effective communication in all the classrooms for both teachers and learners.

Quiz Answer the following questions orally or in writing:

- i. Name three main components in the communication process
- ii. Name three barriers to effective communications
- iii. Is effective communication important in the management and control of classroom (Discuss)

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LESSON 9: SYLLABUS INTERPRETATION AND PROFESSIONAL TOOLS-(Harrison Mulwa)

9.1. Introduction

One of the most central actions in teaching and learning, particularly in a classroom setting, is the process of preparation. In order for the delivery of content to be successful and seamless throughout the year, term or even the 40 minutes of a strand, the facilitator needs to have clear plans of each of the documents. The most common of these documents are the Syllabus, the Scheme of Work, the Strand plan and the record of work.

9.2. Learning outcomes

By the end of this lesson you will be able to:-

- 9.2.1 Define a school syllabus
- 9.2.2 Explain a Scheme of work
- 9.2.3 Explain a strand plan
- 9.2.4 Explain the record of work

9.2.1 The Syllabus

A definition of the Syllabus may be stated as follows:-

A syllabus is a document that shows the content to be covered as well the learning outcomes to be achieved in a given subject within a given period and at a particular level. In Kenya the syllabus is prepared by the KICD (Kenya Institute of Curriculum Development). In order for the Syllabus to be relevant, it has to take into consideration the National Goals of Education, the Learning outcomes of both the subject and the lessons and sub lessons of the subject. ***N.B Lesson here is the topic, according to the Competency based curriculum (CBC), while the lesson is now called the Strand.***

9.2.1 E-tivity on Syllabus

Title	The syllabus
Purpose	If you complete this task, you will be able to explain the syllabus to your peers.
Brief summary of overall Task	To be able to explain the different uses of the syllabus.
Spark	<i>Teaching a subject without a syllabus is like trying to find your way inside a pitch dark room</i>
Individual Contribution	-You are in agreement with the statement here above. 1. Give your reasons after visiting and reading the following link on the syllabus. https://www.education.go.ke/images/GPE_PRIEDE/EGM-Facilitators--Manual.pdf
Interaction Begins	Post your opinions (at least 8 sentences) on <i>the link you will be given during lecture.</i>
E-Moderator interventions	All opinions will be summarized and posted at the end of this lesson
Schedule and Time	This activity should last 2 hours
Next	Read <i>Chapter 5 of Instructional methods in Education</i> for further crucial information on this lesson

9.2.2 The Schemes of Work

A scheme of work is a document that a teacher develops from the curriculum design. A scheme of work shows how the planned curriculum content shall be distributed within the time allocated for the learning area.

A scheme of work helps the teacher to do the following:

- To plan on what resources will be required.
- To decide on the methodology to be used.
- To plan for assessment.

A scheme of work has several components which include the following; week, lesson, strand/theme, sub- strand/sub-theme, specific learning outcomes, suggested learning experiences, suggested learning resources, suggested assessment methods and remarks.

9.2.2 A sample Scheme of Work

School	Grade	Learning Area	Term	Year
<i>Shupavu High</i>	<i>Form 2</i>	<i>The sub-strand</i>	<i>2</i>	<i>2022</i>

Week	Lesson	Strand	Sub strand	Specific learning outcome	Key inquiry question	Learning experiences	Learning resources	Assessment	Reflection
1	1								
	2								
	3								

Adapted from KICD

E-tivity: Scheme of Work Plan.

Title	The Schemes of Work Document
Purpose	If you complete this task you will be able to explain the schemes of work document
Brief Summary of Overall Task	To allow the student to have an in-depth knowledge of the preparation of Schemes of Work
Spark	<i>A wise man plans his journey before undertaking it.</i>
Individual Task	Read the information in this link and answer the question. https://www.education.go.ke/images/GPE_PRIEDE/EGM-Facilitators--Manual.pdf 1.Prepare a scheme of work for one week, for a lesson(topic) in one of your subjects
Discussion begins	After reading from the link, discuss with your peers and post a common answer to the question, in the forum
E-Moderator interventions	At the end of the lesson, the answers will be summarized and posted on the forum
Schedule and Time	This work should take 2 hours

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Next	Read <i>Chapter 5 of <u>Instructional Methods in Education</u></i> for further crucial information on this lesson
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9.2.3 The Strand Plan (Lesson Plan)

A lesson plan is an essential document for effective teaching and learning. A well-done lesson plan helps the teacher to;

- organize the content to be taught in advance focusing clearly on the content to be covered and the way it should be taught hence avoiding vagueness and irrelevance
- plan, prepare and assemble teaching/learning resources
- Present concepts and skills in a systematic manner using appropriate strategies to achieve the stated lesson outcomes
- Manage time well during the lesson
- select and design appropriate assessment methods to evaluate the teaching and learning process

Organization of learning

- This shows where learning will be taking place. It could be in the classroom, or outside the classroom or a visit to a nearby library or farm.

Introduction

- The lesson should be introduced in an interesting and stimulating manner to arouse curiosity in the learners. Integrate the learners' related past experiences as much

Components of the lesson plan

Organization of learning - This shows where learning will be taking place. It could be in the classroom, or outside the classroom or a visit to a nearby library or farm.

Introduction - The lesson should be introduced in an interesting and stimulating manner to arouse curiosity in the learners. Integrate the learners' related past experiences as much as possible, tapping into learners' prior knowledge to prepare them for additional content you're about to introduce.

Lesson development - This is the actual teaching of the Learning area content. The subject matter is divided into steps. Each step should contain one main idea or experience. Explicitly outline how you will present the lesson's concepts to the learners and the activities to be undertaken in each step-in order to achieve the stated outcomes.

It should indicate clearly what and how is to be taught and the learners activities (learning experiences). The teacher should vary the teaching/ learning activities as the need arises.

9.2.4 Sample Strand (Lesson plan)(Adapted from KICD training manual)

School	Learning Area	Class	Date	Time	Roll
Shupavu High School		Form 2	7/3/2020	9:40-10:20	40

Strand:

Sub-Strand:

Specific Learning Outcomes:

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1. By the end of the sub-strand, the learner will be able to:

- a)..... b).....
c).....

2. **Learning resources.** (list here all resources necessary for accomplishment of the specific learning outcomes).

3. **Co-competencies.** (List here all the associated values and PCI s that accompany achievement of the stated learning outcomes)

4. Organization of Learning

i) **Introduction:** Introduce lesson with an appropriate set-induction that will draw the attention of learners to the strand

ii) Lesson(strand)Development

Step 1-Learners react to the set induction

Step 2-Teacher uses learners responses to synthesize the sub-strand using key inquiry questions

Step 3-Teacher delivers the content of the lesson (use *demonstration and other relevant methods*)

Step 4-It is now the learner's turn to apply content and prove to themselves, with teacher guidance that the leaning outcomes have been attained. **Conclusion**

The teacher now gives a relevant activity/assignment to learners.

Reflection. The teacher prepares a self-evaluation on what was achieved/not achieved for the purpose of continuity.

9.2.4. E-activity on Strand (Lesson) Plan

Title	Detailed description of the functioning of the Strand Plan
Purpose	When you read information in the given link, this you will be able to construct an effective lesson plan
Brief Summary of Overall Task	Read the information from the given link and answer the questions below. https://www.education.go.ke/images/GPE_PRIEDE/EGM-Facilitators--Manual.pdf 1. Prepare a 40 minutes lesson plan using the guidelines in the link
Spark	<i>A good sower casts the grain seeds evenly over the entire field.</i> 2.Discuss this saying in relationship to preparation of the strand plan.
Individual Task	Read the following information from the given link. Invite your peers to participate in a discussion of the questions
Interaction Begins	Discuss the questions given, then prepare a common answer and post it on the forum
E-Moderator Interventions	Your answers and opinions will be summarized and given here at the end of the week of the lesson on the forum
Schedule and Time	This task should last 4 hours

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Next	Read Chapter 5 of <i>Instructional methods in Education</i> to get additional information on this lesson
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Record of work

A Record of Work is a document kept by the teacher showing the work that has been done at the end of every lesson, strand or sub-strand. The entries are made daily by the individual teacher. It helps in:

- accountability and transparency of work covered by the teacher
- the continuity of teaching of a class
- that a new teacher has an idea of where to start teaching a class
- the evaluation of schemes of work after a period
- providing uniformity of content coverage in case of several streams

The record tracks the achievement of learning outcomes and the competencies acquired by the learner. The record can be used to show the teacher whether their teaching has been effective in addressing the learning needs of individual learners. It therefore acts as a guide for the teacher to be able to give the required attention to individual learners to ensure the desired outcomes as stated in the curriculum designs are portrayed by all the learners. The progress record can also be used to give the learner and the parents/guardians information about the learner's progress.

Components of a Record of Work

Time Frame: There should be an indication of the date and week when the work was covered.

Work done: strand and sub-strand as derived from the specific learning outcome(s)

Reflection: The remarks column should have statement(s) specifying the success and or challenges of that lesson and recommendations.

Details of The Teacher: include the name, signature or initials of the implementing teacher for accountability.

E-tivity: Record of work

Title	Explaining the Record of Work
Purpose	To know the function of the Record of Work
Brief Summary of Overall Task	Preparation of a Record of Work
Spark	<i>Self-introspection is a virtue</i>
Individual task	Read the section on record of work in this link. 1.From the lesson plan you prepared under that e-tivity, prepare an imaginary record of work, to cover all requirements. https://www.education.go.ke/images/GPE_PRIEDE/EGM-Facilitators--Manual.pdf
Interaction Begins	Invite your peers with whom you shared the lesson plan preparation and do the same for this Record of work exercise.
Moderator interventions	The answers will be summarized before end of the lecture week on the forum.
Schedule	This exercise should take 2 hours

Next	Read <i>chapter 4 of Instructional methods</i> for additional information. Your Lecturer will guide you on how to access this text

Quiz

- What are the different parts of the syllabus?
 - National Goals of Education-Instructional objectives
 - Objectives of the Subject- Subject content in Topics and Sub topics
- In your opinion, why are schemes of Work important to the teacher and the school in general? They help the teacher to plan the lesson/For continuity in fulfilling ed. objectives
- What period of time should a Scheme of Work cover and why Give 5 reasons to explain why it is useful to prepare schemes of work
 - 1 year.-changes in methods, texts, content
 - 1. use time effectively, allow effective assessment of objectives...
 - 4. Give 4 reasons why schemes of work prepared for one group may not be suitable for another group?
 - No group of learners are the same
 - Changes in environment...
 - 5. What are the different parts of a learning outcome in a strand plan?
 - admin details, condition,time,content...
 - 6. 'At the end of the lesson, the learner should understand the causes of the first world war' Rewrite this learning outcome correctly.
...the learner will be able to /give or list the causes.....
- Who benefits from Record of work and why?

The teacher/the in-coming teacher as replacement

 - helps to show areas not fully handled
 - Helps to know highlights of lesson/new inputs/ methods teacher found effective

References

- 1.N. Twoli et al (2007). *Instructional Methods in Education: A Course Book for General Teaching Methods*. K.I.C.D
2. Burden, P. R. & Byrd, D. M (2012). *Methods for Effective Teaching: Meeting the needs of All Students*, 6th edition, Amazon
3. Kolb, D. A. (1984). *Experiential learning: Experience as the source of learning and development* (Vol. 1). Englewood Cliffs, NJ: Prentice-Hall
4. https://www.education.go.ke/images/GPE_PRIEDE/EGM-Facilitators--Manual.pdf

LESSON 10

INCLUSIVE EDUCATION – PROVIDING FOR INDIVIDUAL DIFFERENCES (Supporting ALL Learners)

Dr. Mueni N. Kiio

10.1 Introduction

In every class, there are differences which exist among the learners. The differences could be in terms of their age, gender, ethnic or cultural backgrounds, mental abilities and their academic performance, among others. The teacher needs to be aware of these differences in order to reach every learner at the

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point of his/her educational need. Otherwise, if the differences are not addressed they can affect learners' performance depending on their severity. This lesson addresses some individual differences and how the teacher should handle them in order to create an enabling classroom environment for all the learners.

10.2 Learning outcomes

By the end of the lesson you will be able to:

- 10.2.1 define the term 'Inclusive Education'
- 10.2.2 highlight the main features of Inclusive Education
- 10.2.3 list methods of identifying learners' individual differences
- 10.2.4 give ways of providing for individual differences in your class.

10.2.1 The concept of Inclusive Education

The term Inclusive Education refers to education which accommodates learners with various individual differences. The differences could be in terms of their age, gender, and mental abilities. It is important to be aware of the individual differences which exist among your learners, in order to address them during instruction.

E-tivity 10.2.1 Meaning of Inclusive Education

Title	Meaning of Inclusive Education
Purpose	The purpose of this e-tivity is to enable you to define the concept of Inclusive Education
Brief summary of the overall task	
Spark	
Individual contribution	Explain the concept of Inclusive Education
Interaction begins	1. Post your answer in the discussion forum 2. Read posts from your colleagues 3. Provide constructive feedback on your colleagues' comments.
E-moderator interventions	1. Checking what appears on the platform. 2. Providing feedback on what is posted on the platform. 3. Close the e-tivity.
Schedule and time	The above task should take about one hour.
Next	Types of individual differences among learners

10.2.2 Types of individual differences among learners

There are various types of individual differences which exist among learners in each given class. They include age, gender, mental ability, physical characteristics, physical challenges, emotional stability and economic status. These differences can make learners learn at different rates. Hence the need to identify learners' individual differences in order to reach them at the point of academic needs.

E-tivity 10.2.2 Types of individual differences

Title	Types of individual differences
Purpose	
Brief summary of the overall task	
Spark	
Individual contribution	
Interaction begins	1. Post your answer in the discussion forum 2. Read posts from your colleagues 3. Provide constructive feedback on your colleagues' comments.
E-moderator interventions	1. Checking what appears on the platform. 2. Providing feedback on what is posted on the platform. 3. Close the e-tivity.
Schedule and time	
Next	Methods of identifying learners with individual differences.

10.2.3. Methods of identifying learners with individual differences

This section deals with methods of identifying learners' individual differences or special needs. It is important to know your students well in order to identify those who have special learning needs and to cater to the said needs adequately. You can use some or all the following methods to find out the individual differences which exist among your learners.

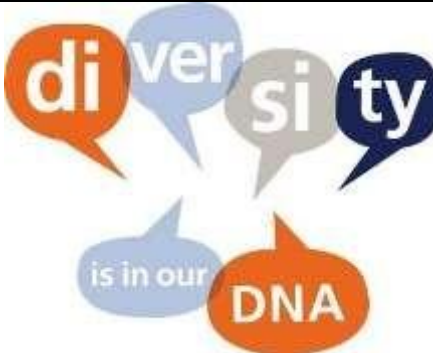
- a) Observation of learners' behaviour in class and outdoor activities.
- b) Use of oral and written interviews.
- c) Use of records, e.g. admission records, medical records, progress records, punishment book, etc.
- d) Use of teacher-learners' conferences.
- e) Use of open days.
- f) Use of tests to gauge learners' understanding of the content covered.

E-tivity 10.2.3 Methods of identifying learners with individual differences

Title	Methods of identifying learners with individual differences
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Purpose	
Brief summary of the overall task	
Spark	
Individual contribution	
Interaction begins	1. Post your answer in the discussion forum 2. Read posts from your colleagues 3. Provide constructive feedback on your colleagues' comments.
E-moderator interventions	1. Checking what appears on the platform. 2. Providing feedback on what is posted on the platform. 3. Close the e-tivity.
Schedule and time	
Next	Methods of providing for individual differences

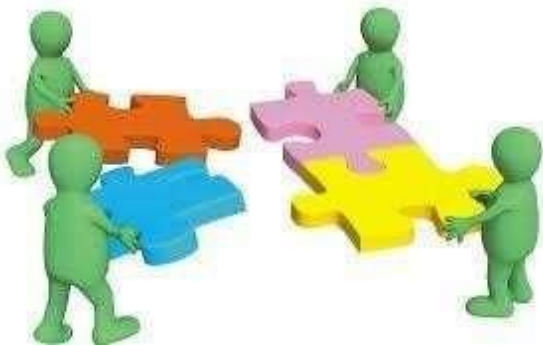
10.2.3 Methods of providing for individual differences

You should note that methods of providing for individual differences will depend on the nature of the differences and time available in a given lesson. They include:

- a) Use of a variety of instructional resources
- b) Use of motivation/encouragement
- c) Use of group work
- d) Use of individualized instruction
- e) Use of remedial work

E-tivity 12.2.3 Methods of providing for individual differences

Title	Methods of providing for individual differences
Purpose	
Brief summary of the overall task	

Spark	
Individual contribution	
Interaction begins	1. Post your answer in the discussion forum 2. Read posts from your colleagues 3. Provide constructive feedback on your colleagues' comments.
E-moderator interventions	1. Checking what appears on the platform. 2. Providing feedback on what is posted on the platform. 3. Close the e-tivity.
Schedule and time	
Next	

10.3 Quiz

- 1) Identify any three individual differences which are likely to be among learners in your class.
- 2) List three ways of identifying learners' individual differences in the class.
- 3) Outline ways of providing for individual differences of learners in your class.
- 4) Discuss challenges which you are likely in your attempt to catering for individual differences among learners in your class.

10.4 References

1. Muthwii, S.M, Ndiku, Kiio, M. (2015) Educational Instruction. Oxford University Press, East Africa Ltd.
2. Kiio M, Munyao L, Manji J. (2005) Explore History and Government Teachers' Guide. Longman, Kenya.
3. Patel, M. & Ayot, H. (1989) Instructional Methods. LIE and KU
4. Twoli N, Maundu J, Muindi D, Kiio M, Kithinji C T. (2007) Instructional Methods in Education: A Course Book for General Methods in Education. KICD, Nairobi.

LESSON 11: COMPETENCY BASED ASSESSMENT AND EVALUATION. John N. Maundu

11.1 Introduction

In teaching and learning, determination of learner performance is a normal practice. At the end of a school term, the classroom teacher usually administers a test to find out how well his/her learners have understood what was taught. Similarly, from time to time, the Ministry of education in Kenya (and

elsewhere) reviews its national curriculum to ensure it is in vogue with contemporary aspirations and practices. These two undertakings, in the school and at the ministry, are examples of assessment and evaluation respectively

11.2 Learning outcomes

By the end of the lesson you will be able to:

- 11.2.1 Distinguish between assessment, measurement and evaluation
- 11.2.2. Distinguish between formative and summative assessment and evaluation
- 11.2.3 Explain the meaning of competency based assessment
- 11.2.4 Explain the meaning of competency based evaluation

11.2.1 Concept of assessment

The term “Assessment” has been used interchangeably with “measurement” and even with “evaluation”. To minimize confusion, assessment has been used in this lesson to refer to various methods and tools used to determine the knowledge and skills, together with the attendant attitudes, learners are acquiring in the course of instruction.

Among the major purposes of assessment include the provision of performance feedback to students, teachers, school administration, parents /guardians and sponsors. Consequently, appropriate steps are taken to facilitate learner achievement at school.

E-tivity 11.2.1 Meaning of Assessment

Title	Meaning of assessment
Purpose	The purpose of this e-tivity is to enable you to construct an acceptable definition of the term ‘Assessment’
Brief summary of overall task	Open the link https://cft.vanderbilt.edu/student-assessment-in-teaching-andlearning/ to learn more about the concept of assessment
Spark	Brighton is a university graduate because he knew the basics of assessment. Learn the tricks to succeed in life.
Individual contribution	In your own words, how would you define the term ‘assessment’?
Interaction begins	Post your definition on the kusoma discussion forum; respond to some of the questions subsequently posted therein about your definition
E-Moderator interventions	• Checking what appears in the discussion platform • Providing feedback on what is posted on the platform
Schedule and time	The task above should take you about 1 hour to complete
Next	Concept of measurement

References

1. Anderson L.W, Krathwohl D.R., Bloom B.S. (2001). A taxonomy for learning, teaching and assessing: A revision of Bloom’s taxonomy of educational objectives. Longman.
2. Secolsky C. Denison D.B. (2012). Handbook on Measurement, Assessment and Evaluation.
3. Black P. & William D. (1998). Assessment and classroom learning
4. Twoli N., Maundu J., Muindi D., Kiio M., Kithinji C. (2007). Instructional Methods in education.KIE.

The concept of measurement

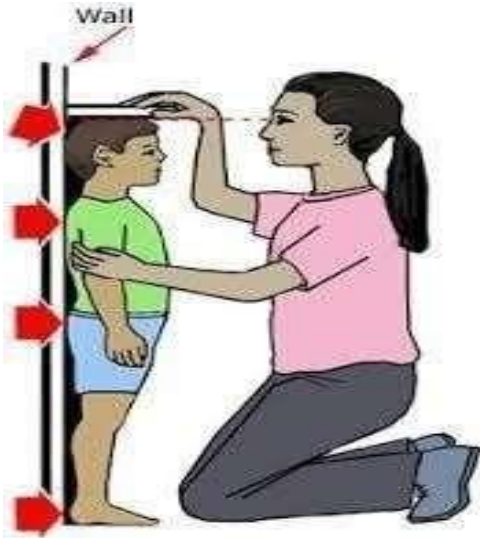
Measurement is a common activity frequently employed by teachers in the course of curriculum implementation. After covering a given topic or set of topics a classroom teacher will usually seek to

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establish what students have understood and how well they have done so. To do this he/she will use tools called tests. Examples of teacher-made test are: Essay, Multiple choice, True/False, Completion, Structured tests and Short answer tests.

E-tivity 11.2.1 Concept of measurement

Title	Concept of measurement
Purpose	The central focus of this e-tivity is to enable you to understand the meaning of measurement and its role in teaching and learning
Summary of overall task	Open this link to get ideas on types of measures used by classroom teachers https://www.geneseo.edu/sites/default/files/sites/education/p12resources-assessmentinforms-teaching.pdf
Spark	 An illustration showing a female teacher in a pink shirt and dark pants kneeling on the floor. She is measuring a young boy's height against a vertical wall. The boy is wearing a green shirt and blue shorts. Red arrows point to the top of the boy's head and the wall. The word 'Wall' is written above the wall line.
Individual contribution	List three types of measurement tools used by classroom teachers
Interaction begins	Post your response in the kusoma discussion forum
E-Moderator interventions	Your answer above should relate to secondary school education in Kenya
Schedule and time	One hour is adequate for the task above
Next	Concept of evaluation

Reference

1. Twoli N., Maundu J., Muindi D., Kiio M., Kithinji C. (2007). Instructional Methods in education.KIE.

Concept of Evaluation

You have learned that one can measure the achievement of learning outcomes through classroom tests. This exercise is called assessment and utilizes measurement to obtain data in the form of scores. In assessment, learners' competencies or abilities are determined. Test scores can be used, in combination with other data, to make decisions on specific aspects of interest in education. The making of a value judgment on the worth, quality, effectiveness or significance of an educational programme, course or

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Unit is referred to as evaluation. In order to pass a comprehensive judgement on the quality and relevance of an educational programme or Unit against stated learning outcomes, it is necessary to obtain information from a variety of stakeholders such as learners, teachers, school administrators, parents/guardians, local leaders and employers .among others. A variety of techniques such as questionnaires, classroom observation and interviews are utilized for such a purpose.

E-tivity 11.2.1 Concept of Evaluation

Title	The meaning of evaluation
Purpose	This e-tivity seeks to enable you to distinguish between assessment and evaluation
Brief summary of overall task	Open the following link to find out the meaning of evaluation https://www.yourarticlelibrary.com/statistics-2/evaluation-in-teaching- and-learning-process-education/92476\
Spark	Students have consistently performed well in this Unit over the last 5 years. I can be among them!
Individual contribution	Explain three types of evaluation covered in the link
Interaction begins	List three key differences between Assessment and evaluation; post them on the kusoma discussion forum and respond to some of the queries that are raised thereafter
E-Moderator interventions	• Providing feedback on differences given by the students • Allowing for comments from the students on my feedback
Schedule and time	It should take you no more than 1 hour to complete the entire task indicated above
Next	Formative Assessment

11.3.1 References

1. Twoli N., Maundu J., Muindi D., Kiio M., Kithinji C. (2007). Instructional Methods in education.KIE.
2. Nicol, D.J., Thomson, A., & Breslin, C. (2014). Rethinking feedback practices in higher education: A peer review perspective. Assessment & Evaluation in Higher Education, 39(1), 102-122.

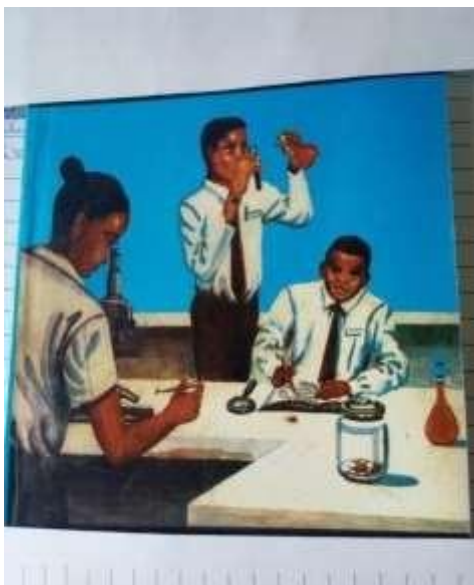
11.2.2 What is formative assessment?

Formative assessment concerns itself with determining the progress and challenges being experienced by both the learners and their respective teachers/lecturers in the course of covering designated content. By analysing the student performance through various tests, interviews and observations of student behaviour, the interested parties can be able to make the necessary changes and modifications in the subject matter, resources and instructional procedures. **E-tivity 11.2.2 Meaning of formative assessment**

Title	Formative assessment
Purpose	Through this activity, you should be able to explain the meaning of formative assessment

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Brief summary of overall task	Watch the video by Michigan department of Education https://www.youtube.com/watch?v=IxW7mk1BGvQ and open the link youtube.com/watch=hkoDWSCGIE4 to understand the concept of Formative assessment
Spark	 What is going on here? Who else, though invisible, is involved?
Individual contribution	In your own words, what is the meaning of formative assessment?
Interaction begins	List three key features of formative assessment and post them on the kusoma discussion forum
E-Moderator interventions	- Assessing the quality of students responses - Providing appropriate feedback to ensure correct understanding
Schedule and time	Plan to spend no more than 40 minutes on the above task
Next	Summative Assessment

11.4.1 References

- Nicol, D., & Macfarlane-Dick, M. Rethinking formative assessment in HE: A Theoretical model and seven principles of good feedback practice. Higher Education Academy:UK. Retrieved from <http://www.heacademy.ac.uk/assets/was%20York%20->
- Shute, V. J. (2008). Focus on formative feedback. Review of Educational Research, 78(1), 153-189.

11.2.2 Meaning of summative assessment

At the end of each education cycle, such as end of Primacy, Secondary school programme or end of a college/ university semester, it is common practice to determine learner performance. This is usually done through national and college/university examinations. Such an activity seeks to find out the kinds

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of knowledge and skills, as well as the attendant attitudes that learners will have acquired. Such an exercise falls in the realm of summative assessment.

E-tivity 11.5.1: What is summative assessment?

Title	Meaning of summative assessment
Purpose	This activity will enable you to explain the concept of summative assessment
Brief summary of overall task	After watching the video in the links youtube.com/watch?v=xgwSZVMOjGU and https://study.com/academy/lesson/summative-assessments-examples-typesquiz.html you will be able to explain the meaning of summative assessment
Spark	How am I academically and socially different now that I have completed this ECT 201 Unit?
Individual contribution	Critically discuss, in approximately 300 words, whether summative assessment is a fair judgement of a student's academic performance
Interaction begins	List 2 two reasons why summative assessment is not a comprehensive evaluation of a student's acquired potential and 2 reasons why it is. Post your response on the kusoma platform
E-Moderator interventions	Critically examine and provide feedback on a sample of the students understanding of summative assessment
Schedule and time	Plan to spend a total of not more than two hours on the above exercises
Next	Competency based assessment

References <https://www.aeseducation.com/blog/>

11.2.3 Meaning of Competency based Assessment (CBA)

When one is hired to do a particular task or job, the quality of his/her output or performance will indicate his/her ability to perform the given task. Further, the time taken to complete the assignment will be a measure of competency. In Education too, our competencies are usually judged via various assigned practical tasks or activities; for the classroom teacher/lecturer, this is done through specific assessment engagements, mainly quizzes, examinations or projects. Through these activities, the skills or competencies acquired by each learner can be identified both during the process of curriculum implementation and upon completion of a given programme of study. The former stage is referred to as competency based assessment (CBA) while the latter is in the realm of competency based evaluation (CBE).

E-tivity 11.2.3 Concept of CBA

Title	What is Competency based assessment?
Purpose	By the end of this e-tivity, you will be able, using examples, to explain the meaning and purpose of CBA
Brief summary of overall task	The video links https://www.youtube.com/watch?v=IXKSU6btG6s and https://www.youtube.com/watch?v=YhF-Z6xSXs8 will enable you to explain the essence of CBA
Spark	Wow! Am able to perform this activity!
Individual contribution	1. Explain, in your own words, the basic meaning of CBA as used in teaching and learning 2. Identify an academic activity you enjoy doing competently

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Interaction begins	State two purposes that CBA serves in instruction and post your answer in the kusoma discussion forum
E-Moderator interventions	Your responses should focus on the teaching and learning environment of formal schooling
Schedule and time	Spend a maximum of 40 minutes in the exercises above
Next	Competency based evaluation

References

1. <https://medium.com/@ejhudson/how-to-design-a-competency-based-assessment-39f312235bde>

11.2.4 Concept of Competency based Evaluation (CBE)

As noted earlier, Evaluation is an activity that seeks to determine the value of an educational programme or process. Therefore, CBE is an undertaking that focuses on the kinds of abilities or competencies that learners have acquired over a definite period of schooling such as at the end of 4 years at secondary school or university. The judgement about learner competencies will be made against specific criteria such as the learning outcomes stipulated for the course content in question.

E-tivity 11.2.4 the concept of CBE

Title	Meaning of CBE
Purpose	This e-tivity will enable you to explain the essentials of competency based evaluation that education evaluators should consider at both formative and summative level
Brief summary of overall task	Through https://www.youtube.com/watch?v=FROloaXJldw you will learn the basic requirements of the 21st century skills and relate them to both the formative and summative level of competency based assessment and evaluation; https://www.rasmussen.edu/student-experience/college-life/what-is-competency-based-education/ provides some background on online competency education; in https://files.eric.ed.gov/fulltext/EJ1167600.pdf competency based evaluation in higher education is discussed. These two latter sources will enable you to start thinking about the need for you to develop competencies that will facilitate you to function effectively in the world today
Spark	ECT 201 will enrich me with 21st century skills!
Individual contribution	In one sentence, explain the difference between competency based assessment and competency based evaluation
Interaction begins	• List the four key skills in the 1st link • Provide two other links that give information about competency based evaluation
E-Moderator interventions	Ensure your responses relate to education in general and to teaching and learning in particular.

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Schedule and time	One and a half hours should be adequate for the tasks given above
Next	Innovative teaching strategies

References

1. <https://www.mooc-list.com/course/assessment-and-teaching-21st-century-skills-coursera>
2. *The Journal of Competency-Based Education*
3. *International Education Studies; Vol. 11, No. 2; 2018 ISSN 1913-9020 E-ISSN 1913-9039*
Published by Canadian Centre of Science and Education

Assessment questions

1. List two administrative purposes that assessment serves in teaching and learning
2. Distinguish between measurement and assessment
3. Who is best placed to undertake formative assessment and why?
4. State and briefly explain **one** value of the KCSE examinations in one of your teaching subjects
5. Design an activity to determine how well your classmates can communicate verbal messages
6. State any **three** key factors that educational evaluators attempting to identify learning competencies among students should consider

Core Reading Material for the Course

Burden, P. R. & Byrd, D. M (2012). *Methods for Effective Teaching: Meeting the needs of All Students*, 6th edition, Amazon

Kolb, D. A. (1984). *Experiential learning: Experience as the source of learning and development* (Vol. 1). Englewood Cliffs, NJ: Prentice-Hall.

N. Twoli et al (2007). *Instructional Methods in Education: A Course Book for General Teaching Methods*.

K.I.C.D

Olga, Bernard- Cavero, Núria Llevot (2018) *New Pedagogical Challenges in the 21st Century: Contributions of Research in Education*.

Stephen Leder 2018 *Transformative Pedagogic Practice: Education for sustainable development and*

Yehudith Weinberger, Zipora Libman (2018) *Contemporary Pedagogies in Teacher Education and Development*

Journals

Australasian Journal of Educational Technology

Educational Technology Research and Development

Educational Technology

British Journal of Educational Technology

Cognition and Instruction

e- Materials

<https://www.springer.com/gp/book/9780792366621>

<https://www.jstor.org/action/showAdvancedSearch>

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+254704597085

Recommended Reading Material

Borich, G. D. (2013). Effective Teaching Methods, Research-based Practice, 8th edition, Amazon

Etisi, M. Z. (2015). Effective Teaching Practices: Empowering Students to Succeed, University of Nairobi Press

Kumar, P. J. (2006). Modern Methods of teaching and learning, Rajat Publication, New Delhi.

Mukwa, C. W & Too, J. K. (2002). General Instructional Methods. Moi University Press, Eldoret: Kenya

Patel M and Ayot (1989). Instructional Methods. LIE and K.U: ODA Projects: Further Reading Texts: Teaching in further education; an outline of Principals and Practice.

e- Materials

<https://www.springer.com/gp/book/9780792366621>

<https://www.jstor.org/action/showAdvancedSearch>

ANSWERS TO ASSESSMENT QUESTIONS

LESSON 1

- Assessment Questions**
1. Differentiate between a goal and a learning outcome
 2. State the other four goals of education apart from the four given in the notes,
 3. Highlight key features in the relationship between teaching and learning.
 4. Explain the following components of teaching:

a) The content b) The methods of instructions

ANSWERS

1 Goals are broad and general and are concerned with the intention of a specific system of education while learning outcomes are narrow and specific and they are limited to the expected lesson.

2. Promote social equality and social responsibility.

- Promote respect for and development of Kenya's rich and varied cultures.

- Promote international consciousness and foster positive attitudes towards other nations. -

- Promote positive attitudes towards good health and environmental protection.

3. Interrelatedness

- Interconnectedness

- Interdependence

- Flexibility

- Interactive

4 a) – is the subject matter to be taught.

b) The way the teacher is going to teach a particular lesson. Is he /she going to use teacher oriented method or learner-centred.

LESSON 2

Assessment Questions:

1. What are the steps of the instructional process or the parts of a systematic instruction?

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2. Answer: The steps of the instructional process or the parts of a systematic instruction are the following: a. Define objectives b. Chose appropriate methods c. Select materials, equipment and facilities d. Assign personal roles e. Implement the instruction f. Evaluate outcomes g. Refine the process
3. *What does each step mean?*
Answer: Each step means that: a. Define objectives- Instruction begins with the definition of instructional objectives that consider the students' needs, interest, and readiness. b. Chose appropriate methods- On the basis of this objective, the teacher selects the appropriate teaching methods to be used.
4. *Are the elements or phases of instruction in the systems approach design independent of one another or do they relate to one another?*
Answer: All steps in the system approach to instruction are interrelated with each other. It is an orchestrated learning pattern with all parts harmoniously integrated into the whole. As a teacher, you must do each step in order to have a better learning process and the students or the learner must be your main focus.

LESSON 3

- 1) What can you do to ensure success of a laboratory experiment?
-preview experiment
-give instructions on how to proceed
- assess results after experiment
- 2) Heuristic methods are useful since they involve the learner. Discuss.
-There is deep learning here. The learner is answering on the why and wherefore of facts.learner has a chance to think, reflect on ideas thus being in charge of his own learning.
- 3) The lecture method is best for covering a lot of content in any given field. What are possible drawbacks to this assertion?
-The learner is a passive listener
-The learner has little time to reflect and digest on what he is learning
- 4) 'When I do a demo in class, I am assured that learners have understood everything' Comment on this teacher's report.
-The learners may not have understood all since they did not do the action themselves. They needed to be allowed to have hands -on experience for the teacher to be fully sure learning has taken place.
5. Think of 2 areas of any of your teaching subjects and say how you can apply experiential teaching to improve learning.
Consider how you would guide your learners through these steps in order to concretise the learning; Experience----reflection on the experience-----Conceptualisation.....Experiment to prove the first three steps.

LESSON 4

1. Compare teachers and the students' roles in traditional learning approaches and the competence based teaching methods

A Competence based Educational approach allows learners to progress at their own pace whereas Traditional learning approaches require all learners in a class to progress at the same rate.(CORRECT)

- B Competence based Educational approach and Traditional learning approaches differ in the use of instructional media only.
 - C Problem solving and collaborative learning methods are examples of Competence based Educational and the Traditional learning approaches
 - D Competence based Educational approach are meant for small classes whereas Traditional learning approaches are best for large classes.
2. Which of the following statements is correct about collaborative and the cooperative learning methods.
- A Cooperative learning is a specific kind of collaborative learning where the individual contribution is the focus.
 - B Collaborative learning method is also known as cooperative learning method.
 - C Collaborative learning method focuses at the team work whereas in cooperative learning there is no team work
 - D Both learning methods are about groups of students working together towards a common goal. (CORRECT)

LESSON 5

1. Which of the following statements gives the elements of computational learning?
- A. A problem and its systematic solution (CORRECT)
 - B A problem and a solution that both humans and computers can understand.
 - C Pattern recognition, abstraction and decomposition
 - D Algorithm thinking, Decomposition, pattern recognition and abstraction
2. Why is learning by doing an important learning method?
- A Students make some sense of their learning experiences
 - B Learners are engaged in hands on learning activities that leads to acquisition of practical skills. (CORRECT)
 - C Learners can effectively acquire knowledge without teacher involvement.
 - D Students can write a project report.

LESSON 6:

ODHIAMBO DARREN GREGORY
+254704597085

ANSWERS

1. B
2. Collaborative learning means being able to acquire knowledge, skills and attitudes through social interaction in a friendly, give-and – take environment
3. C
4. D
5. a) Three advantages of teaching large classes: learning and using new technology-based instructional technologies, sharing information to a large group of learners, call for creativity in designing content appropriate for large number of learners
b) Three limitations of teaching large classes: adequacy of resources is a challenge, personal experience by each learner in the use of materials by all learners not guaranteed, marking of work submitted by all students problematic

LESSON 7

ANSWERS

1. Intrinsic motivation is when a learner is propelled to learn or perform a particular task because of the pleasure he/she derives from the process or the activity .On the other hand extrinsic motivation is when the incentive to learn or perform is influenced by outside forces such as the rewards they will receive for successful performance of an activity.
2. That for a child to learn which is a higher need we must first satisfy the needs below pursuit of education.
 - Carter for basic needs of the learner e.g food, water
 - Ensure they are secure in school and in the classroom before learning. This includes emotional security as well.
 - The teacher has to show love to the students unconditional .
 - The learner must be loved and respected.
 - Teacher has to create supportive learning environment
 - When the above three are met then it will be very easy for learning and teaching to take place.
3. That in the classroom learning can take place through ;earners observing the teacher . The teacher becom es a role model.
 - They also learn through observing the behavior of fellow students.
 - Teach them to share their knowledge , resources through discussions.
 - That the teachers should use observation/ experimental methods of teaching to impact on learners.
4. Inspire him/her to aim higher to achieve their dreams
 - Provide new learning experiences
 - Give them the power of choice
 - Encourage them to collaborate with other learners within the class, the school and even with other schools.
 - Focus on their personam interests

5. –employ variety of methods of teaching
 - use reinforcement technique
 - vary the stimulus
 - maintain success expectations
 - in cooperate learning resources

LESSON 8 Answers

- i. Source, Message, Receiver
- ii. Physical, Psychological, Language differences
- iii. Effective communication is very important in management and control in classroom.

LESSON 9

1. What are the different parts of the syllabus?

- National Goals of Education-Instructional objectives
- Objectives of the Subject- Subject content in Topics and Sub topics

2. In your opinion, why are schemes of Work important to the teacher and the school in general? They help the teacher to plan the lesson/For continuity in fulfilling ed. objectives

3. What period of time should a Scheme of Work cover and why Give 5 reasons to explain why it is useful to prepare schemes of work

-1 year.-changes in methods, texts, content

1. use time effectively, allow effective assessment of objectives...

4. Give 4 reasons why schemes of work prepared for one group may not be suitable for another group?

-No group of learners are the same

-Changes in environment...

5. What are the different parts of a learning outcome in a strand plan?

-admin details, condition, time, content...

6. 'At the end of the lesson, the learner should understand the causes of the first world war' Rewrite this learning outcome correctly.

...the learner will be able to /give or list the causes.....

7. Who benefits from Record of work and why?

The teacher/the in-coming teacher as replacement

-helps to show areas not fully handled

-Helps to know highlights of lesson/new inputs/ methods teacher found effective

LESSON 10

1. Socio-economic; Intellectual ability; Language and cultural diversity; Physical and emotional

2. Referral facilities; observations; interviews,

- Use of a variety of instructional resources
- Use of motivation/encouragement
- Use of group work
- Use of individualized instruction □ Use of remedial work.

4. Rebellion; shyness, non-cooperation, lack of facilities

LESSON 11

a) Two administrative purposes of assessment: Facilitates guidance and counselling services for learners;

Enables decision making regarding promotion of learners to the next academic level

1. Assessment is a process that deals with determination of learner performance including what they should do in order to improve; measurement is the use of specific tools (such as tests) to obtain quantitative and qualitative information about learner achievement
2. The classroom teacher, being the implementer of the curriculum, is best placed to determine learner progress in the course of teaching and learning.
3. All school subjects, in one way or another, serve to provide cognitive growth to learners as measured by the KCSE examinations
4. The activity presented should facilitate verbal communication, be relevant and interesting to learners, stimulate questioning, provide an opportunity to state and explain information clearly. It should also enable learners to seek for clarification of ideas and adherence to appropriate use of grammar and vocabulary. Examples will vary according to areas of study.
5. Key factors to be considered in CBA: Learner's ability to use **Learning skills** (creativity, critical thinking, communication and collaboration), **Literacy skills** (ability to source genuine information from known outlets including the technology used in publishing the content) and **Life skills** (social interaction abilities-negotiation, respect for opinion, patience etc.)